The Confounding Frontier: Exact Spectral Bias Laws, Calibrated Detection, and the Limits of Deconfounding under Dense Hidden Confounding

Hugo Souto*

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Abstract

For high-dimensional regression with dense latent confounding in the proportional regime $c = p/n$, we prove an exact capture law: minimum-norm least squares inherits along each spike direction q_j a bias component rescaled by $\mathrm{cap}_j = (1 + l_j)/(c + l_j)$. The law follows from an elementary overlap-free argument, extends to any fixed rank, interpolates through ridge regression, and yields a companion law under which hard-trim-then-regress treatment estimates stay stable across c while ordinary least squares inflates roughly six-fold. Building on the same null theory, we construct a calibrated spike-coordinate alarm whose realized power frontier matches its deterministic prediction with median empirical-to-predicted ratio 1.02 in simulation strata. On detection limits, we prove a population detachment boundary for augmented-spectrum alarms, show that a stronger universal impossibility claim fails under pre-registered tests of our own design, scope every blindness statement to named statistic classes, and derive an exact visibility law that locates the born-blind crossing of quadratic probes at the Marchenko–Pastur point $c = 1$ for the scaling family studied. On estimation, we prove that within the class of spectral-weight estimators no soft member matches the transmission floor of hard trimming on subcritical cells, consistent with the empirical failure of tuned soft methods throughout the harmful region, including exact collapse to ordinary least squares on roughly a third of harmful cells. Semi-synthetic benchmarks on three real covariate geometries and a released package complete the evidence.

1 Introduction

Imagine a scientist who regresses a response on p covariates observed over n units, with p comparable to, or larger than, n: several thousand gene expressions, hundreds of firm or asset characteristics, brain-imaging summaries, text embeddings. The covariates were not randomized, so it is plausible that a handful of unmeasured common causes move many covariates and the response simultaneously, a situation we call dense latent confounding. Three questions follow. First, is the confounding biasing my estimates, and by how much? Second, can I tell from the observed data alone that it is there? Third, if I adjust spectrally, for instance by trimming leading principal components, does the adjustment help me or hurt me? Common practice answers the second question by inspecting the scree plot of the sample eigenvalues. A recurring message of this paper is that this practice fails

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quantitatively in both directions: scree-type evidence fires on ordinary design structure whether or not confounding is present, and it stays silent exactly where silence is dangerous.

We study the standard latent-factor model for one observational dataset,

$$X = \Lambda f + u, \qquad Y = \beta' X + \gamma' f + \varepsilon,$$

where X collects the p covariates of a generic unit, Y is the scalar response, f stacks r hidden factors with r fixed, u is covariate noise with variance σ_u^2, and ε is response noise; only (Y, X) is observed. Here Λ is a dense $p \times r$ loading matrix, $\beta \in \mathrm{R}^p$ is the structural coefficient vector the scientist wants, and $\gamma \in \mathrm{R}^r$ is the confounder-response link. The aspect ratio is $c = p/n$, and $\Sigma_X = \sigma_u^2 I + \Lambda\Lambda'$ is the population covariance of X. The spike strengths $l_1 \geq \cdots \geq l_r \geq 0$ are the eigenvalues of $\Lambda\Lambda'/\sigma_u^2$, so the population eigenvalues of Σ_X/σ_u^2 exceed one exactly by the l_j. Informally, l_j measures how far the j-th hidden factor stands above the noise floor, and $\sqrt{c}$ is the visibility threshold inherited from random matrix theory: a factor with $l_j < \sqrt{c}$ leaves no outlier eigenvalue, while one with $l_j > \sqrt{c}$ produces a separated spike. Two objects summarize the confounding itself: the bias direction $b = \Lambda\gamma$, the p-vector through which hidden factors enter the conditional mean of Y, and $g = \|\gamma\|_2$, the overall confounding strength; the alignment angle θ between γ and the dominant factor direction determines where the confounding mass rides.

The geometry behind our results is a decoupling. Visibility depends only on the spectral profile (l_j, c), because eigenvalue outliers do not see γ. Harmfulness depends on $b = \Lambda\gamma$. Since these are different functions of the parameters, there are regions of (l, c, g, θ) where the confounding is invisible to any eigenvalue inspection yet biases ordinary least squares by a large constant, and regions where a spectacular spike is visible yet essentially harmless. A practitioner equipped only with a scree plot cannot tell these worlds apart.

Stated in one sentence, this paper contributes exact spectral theory for hidden dense confounding: a validated capture law for the bias confounding induces, a calibrated detection test whose power frontier is predictable to within a factor of about one, a proof-backed map of what cannot be detected and by which classes of statistics, and a trim-then-regress recipe for treatment effects, with the failure of tuned soft-trim estimators characterized rather than hidden. In detail:

1. *An exact capture law for confounding bias.* For ordinary least squares at aspect ratios $c > 1$, the confounding-induced bias along the j-th spike direction is capped exactly by $\mathrm{cap}_j = (1 + l_j)/(c + l_j)$, and at $c \leq 1$ an exact identity pins the bias down completely. The law is proved elementarily at $r = 1$ (a corrected channel assembly plus a Haar-probe symmetry argument, with no local-law input) and reduced at fixed r through Woodbury identities and vanishing lemmas. A companion ridge formula interpolates the cap continuously in the regularization parameter and collapses, as the regularization vanishes, to the min-norm OLS law; a trimmed-τ law extends the picture to scalar treatment effects, predicting the OLS attenuation denominator excess $\Lambda_D = 1.558$ against a measured 1.5548 ± 0.072. Simulation overlays confirm the theory: 91.7% of the main-grid cells fall within 10% of the predicted bias at $n = 2000$ and every cell does so at the neighboring tiers, with ridge deviations at median 0.0%.

2. *A calibrated detection alarm with a predictable power frontier.* Our statistic S2, a calibrated maximum of standardized spike coordinates, ships with a finite-sample null law: its Monte Carlo 95th percentile exceeds the naive Gaussian-maximum scale by a factor of about 1.7, which a noise-aware recalibration captures (pooled chi-square $p = 0.41$; 96.8% of null cells inside the $|z| \leq 2$ band). Its predicted power frontier matches simulation: the median ratio of empirical to predicted 80%-power strengths is 1.02, with stratum ratios 0.93, 1.25, and 1.02, all within the predeclared factor 1.5.

3. *A map of what cannot be detected.* We prove a population detachment boundary for augmented-spectrum alarms (Proposition 4): for subcritical dense confounding under response standardization no eigenvalue detaches from the bulk at any strength, the low-edge wake criterion fails at every studied aspect ratio, and the limiting bulk spectra under null and alternative coincide. A stronger universal impossibility in the style of Onatski–Moreira–Hallin (OMH) (Onatski et al., 2013) was planned, pre-registered, and destroyed by our own falsifiers: the true augmented top eigenvalue separates consistently below the boundary through finite-sample fluctuations that leave the bulk laws untouched, with geometry-dependent direction, and a pipeline erratum behind the frozen S1 statistic explains part of how the wrong picture survived. What ships is layered: blindness is trivial for purely covariate-based alarms, proved for the spike-coordinate calibrated alarm we ship (power ceiling confirmed in nine of nine blind strata), and false for the augmented top eigenvalue. Complementing this map, an exact visibility law for quadratic probes exhibits satiation and places the born-blind crossing of our scaling calibration exactly at $c = 1$; learning-based probes confirm operational invisibility at $c = 0.8$ through $g \approx 0.8$ while detecting readily at $c = 0.2$ (AUC up to 1.0), so probe-level invisibility is claimed only in intermediate-to-high-c regions.

4. *A characterized negative result for soft tuning, and a recipe that survives.* Empirical-Bayes-tuned soft spectral trimming, the natural adaptive member of the deconfounding family, collapses to OLS or loses to hard trimming in every harmful cell of our grids: its no-regret share (the share of cells with mean-bias norm within 1.05× the best baseline's) is 4.1% against a predeclared requirement of 95%. A collapse lemma explains part of the failure mechanically (the empirical-Bayes ratio vanishes under our primary ledger, driving the soft weights to one, so the tuned estimator reproduces OLS identically in 31 of the 97 harmful cells), and a dominance theorem explains why Onatski hard trimming wins outright (best baseline in 89 of 97 harmful cells). What survives is trim-then-regress: treatment error stays flat in the aspect ratio under hard trimming while OLS error inflates about six-fold, and on real geometry the trimmed estimate is 0.173 against OLS 0.421 at a true treatment effect of 1 (ratio 2.43, correct sign in 100% of replicates).

5. *Semi-synthetic benchmarks and a released package.* We inject known dense confounding into three real covariate geometries (AddNeuroMed gene expression, a featurized IHDP covariate set, and the k401ksubs wealth panel) across six configurations. Before injection, every configuration is massively outlier-positive and the scree alarm rejects always (Tracy-Widom statistics up to 48,431), including under matched nulls and permutation, confirming that spectrum inspection carries no information about confounding. After injection, alarm power straddles each family's own predicted frontier (power between 0.14 and 0.21 at half the predicted strength, and between 0.975 and 1.00 at twice it) while size stays between 0.000 and 0.003 under every negative control across roughly 3,000 control evaluations; incumbent confounding-strength diagnostics track injected strength monotonically but turn anti-conservative out of sample (rejection rates 0.07–0.92 and 0.61–1.00). The pipeline ships as the `confounderalarm` package (v0.1.0), which returns a verdict, a p-value, estimates of (r, l, c), frontier placement, a blind-region certificate, and a recommended Onatski-trim adjustment; every benchmark cell replicates to machine precision (largest statistic shift 7.7×10^{-13}) on independent hardware.

The remainder of the paper is organized as follows. Section 2 fixes the models, normalizations, estimands, and the assumption ledger, including the identification subtlety that only a projection of β is determined by second moments. Section 3 states and proves the capture law, its ridge interpolation, and the trimmed-τ law for treatment effects. Section 4 constructs the calibrated alarm and compares measured power against the predicted frontier. Section 5 maps what cannot be detected: a proved detachment boundary, a falsified universal impossibility reported as an instructive negative result together with its pipeline erratum, the class-scoped blindness of the alarm we ship, and the quadratic-probe visibility boundary. Section 6 reports the estimation negative-result analysis: why oracle-tuned soft trimming fails, why hard trimming dominates, and what survives. Section 7 validates the toolkit on the semi-synthetic benchmarks and describes the package. Section 8 collects limitations, and Section 9 discusses how our tests complement, rather than replace, sensitivity-analysis practice.

1.1 Related work

Closest to us is the spectral deconfounding family. Cevid et al. (2020) showed that dense hidden confounding destroys consistency of sparse high-dimensional regression and that suitable spectral transformations of the design restore it. The idea has been extended to sparse additive models by Scheidegger et al. (2025), carried into random forests through a spectral loss by Ulmer et al. (2025), and combined with boosting and an empirical-Bayes tuning of the spectral loss by Nava et al. (2026). These are estimation-method papers: they deliver rates and consistency guarantees, often with arbitrary or heuristic default tuning, and none of them contains a bias phase diagram, a detectability or removability frontier, a calibrated test with controlled size, or an impossibility statement. Our optimality verdict is positioned squarely against this family, in particular against the empirical-Bayes tuning of Nava et al. (2026), and Section 6 shows that tuned soft weights lose comprehensively to hard trimming on harmful cells.

A second line quantifies confounding strength. Janzing and Schölkopf (2018) estimate the degree of confounding from an independence-of-mechanisms principle via spectral concentration arguments, and Rendsburg et al. (2024) establish consistency of a random-matrix-corrected version of that estimator. This line stops at description: it ships no exact null distribution, no law for the induced bias, no statement about when confounding matters for a specific causal estimand, and no guidance about removal.

A third neighborhood is high-dimensional endogeneity and latent-factor inference. Fan and Liao (2014) construct factor-adjusted projected statistics for testing under strong factors, Guo et al. (2022) provide debiased inference for sparse coefficients under hidden confounding, and Wang and Blei (2019) develop the multiple-causes identification argument. Their targets are significance, interval estimation, and identification; none of them analyzes the spectral phase behavior of bias, detectability, or removability, which is the object here.

Methodologically we compose the random matrix toolbox into causal-estimand functionals: the Marchenko-Pastur law (Marchenko and Pastur, 1967), the Baik-Ben Arous-Peché transition (Baik et al., 2005), Tracy-Widom flank distributions (Johnstone, 2001), outlier locations and eigenvector overlaps (Benaych-Georges and Nadakuditi, 2011, 2012), sharp power analysis of spectral tests near the transition (Onatski et al., 2013), consistent estimation of the number of spikes (Onatski, 2010), exact ridge risk (Dobriban and Wager, 2018; Hastie et al., 2022), and anisotropic local laws (Knowles and Yin, 2017). Our contribution is not any single one of these laws but their assembly into bias caps for OLS, ridge, and spectral trims, and into power frontiers for confounding alarms.

To our knowledge, no prior work provides the combination delivered here: a bias phase diagram, a detectability frontier with a structural removability verdict, a calibrated alarm, and an impossibility map stating which classes of statistics are blind where. Targeted vocabulary searches support the gap claim: arXiv API queries executed on 2026-08-23 returned zero hits for “deconfounding” AND “phase transition”, zero for “hidden confounding” AND “random matrix”, zero each for “confounding” paired with “Benaych-Georges” or “Tracy-Widom”, and zero for “confounder detection” AND "spike", while "spectral deconfounding" returns exactly the four-paper family above; a forward-citation screen of the family's records (65 unique works merged across two APIs) surfaced no phase-transition or frontier analysis either.

2 Model, estimands, and assumption ledger

This section fixes the models, normalizations, and estimands, records the identification subtlety behind the assumption ledger, and states the detection problem.

2.1 Data-generating models

M1: vector structural target. One observational dataset consists of n independent copies of $(f_i, u_i, \varepsilon_i)$, with f_i an r-vector of latent factors, u_i a p-vector of design noise, and scalar ε_i, generating

$$X = \Lambda f + u, \qquad Y = \beta^\top X + \gamma^\top f + \varepsilon.$$

Here Λ collects $p \times r$ dense loadings, β is the structural coefficient vector, and γ couples factors to response; the observable data are (Y, X) only, with f latent. All confounding enters the conditional mean through the single p-vector $b := \Lambda\gamma$, the bias direction; this form is second-moment equivalent, under Gaussian errors, to a joint-Gaussian model with $\mathrm{Cov}(X, Y) = \Sigma_X \beta + \Lambda\gamma$.

M2: scalar treatment target. M2 adds an observed treatment D,

$$D = \pi^\top X + \delta^\top f + \nu, \qquad Y = \tau D + \beta^\top X + \gamma^\top f + \varepsilon,$$

with independent noise ν and scalar target τ, reserved for the treatment-effect analysis; all prior development uses M1.

Binding normalizations. Six conventions hold throughout. First, $\sigma_u^2 = 1$ unless stated otherwise, and spike strengths are relative to σ_u^2, so statements are scale equivariant. Second, $l_1 \geq \cdots \geq l_r \geq 0$ are the eigenvalues of $\Lambda\Lambda^\top/\sigma_u^2$, so the population eigenvalues of Σ_X/σ_u^2 are $1 + l_j$, where $\Sigma_X = \mathrm{Var}(X) = \sigma_u^2 I + \Lambda\Lambda^\top$. Third, $c = p/n$ may take any positive value. Fourth, $\|\beta\|_2 = 1$ and $\gamma = g\,\mathrm{dir}(\theta)$ with $g = \|\gamma\|_2$ and $\mathrm{dir}(\theta) = \cos(\theta)e_1 + \sin(\theta)e_2$ in the coordinate basis of factor space. Fifth, $\Lambda = Q\,\mathrm{diag}(\sigma_u\sqrt{l_1}, \ldots, \sigma_u\sqrt{l_r})$ with Haar-isotropically drawn orthonormal columns Q; dense loadings means exactly this isotropic row distribution. Sixth, u_{ij} and ε_i are standard normal at finite n.

Why this geometry. Write $u_j := Qe_j$ for the population eigenvector directions of Σ_X, with eigenvalues $1 + l_j$. Since $b = \sum_{j \leq r} \sigma_u \sqrt{l_j}\, \gamma_j u_j$, the angle θ controls how much confounding mass rides on strong versus weak factor directions while leaving the design spectrum $(l_1, \ldots, l_r)$, hence its scree and Tracy–Widom appearance, untouched. θ thus slides confounding between directions an eigenvalue alarm can see and directions it cannot; separating visible from harmful is why θ grids every experiment below.

One caveat is recorded honestly. Exactly zero OLS bias together with nonzero spikes requires $\Lambda\gamma = 0$, which cannot occur for generic full-rank Λ; "harmless" therefore always means small bias ratio, operationalized as γ aligned with negligible-strength factors (auxiliary cells take secondary spikes $l = 10^{-4}$, whose per-direction OLS bias coefficient $\sqrt{l}/(1+l)$ is near 0.01), never exactly zero.

2.2 Estimands

Definition 1 (Loading-conditional bias)**.** For an estimator T of β under M1, define the bias functional as the conditional mean of the estimation error,

$$\mathrm{Bias}(T) := \mathbb{E}\big[\, T(Y,X) - \beta \,\big|\, Q, l \,\big],$$

where the expectation integrates out f, u, ε, and the independent A4a draw of β, while the realized loading geometry (Q,l) is held fixed. $\mathrm{Bias}(T)$ is thus a deterministic vector attached to the geometry, not a random quantity. The scalar gate summary is its norm relative to the signal, $\mathrm{rel\_bias}(T) = \|\mathrm{Bias}(T)\|_2/\|\beta\|_2$ with $\|\beta\|_2 = 1$, and every "mean-bias norm" below means exactly this number.

The conditioning carries content: were Q redrawn Haar-isotropically on every replicate, the unconditional mean bias vector would vanish by spherical symmetry, while each dataset still suffers an $O(1)$ bias along its own realized spike directions, so the unconditional moment averages away exactly the error practitioners incur. This functional is what simulations estimate (Q drawn once per configuration) and what the laws of Section 3 predict.

A companion functional isolates what confounding adds over pure fitting artifacts:

$$\mathrm{rel\_bias}_{\mathrm{conf}}(T) = \frac{\big\|\mathbb{E}[T \mid \Lambda, \gamma] - \mathbb{E}[T \mid \Lambda, \gamma = 0]\big\|_2}{\|\beta\|_2},$$

estimated from twin arms under common random seeds. The subtraction matters for $c > 1$, where minimum-norm least squares shrinks β even without confounding, so twin arms are run on that side. At $c \le 1$ an exact finite-n identity (Section 3) makes the two functionals coincide, and the plain bias functional is used directly with no twin arm.

Definition 2 (Outlier visibility)**.** $V(X) = \lambda_{\max}(X^\top X/n)/\sigma_u^2$, compared against the Tracy–Widom upper 99% quantile for white noise scaled by $1/n$; configurations exceeding it are outlier-positive.

The detection frontier organizes the empirical questions; a removal frontier was also registered, and we state at once how it is discharged. The detection frontier is $s_{\mathrm{detect}}(l,c)$, the minimal effective detection spike (Definition 3) at which the calibrated alarm of Section 4 reaches power 0.8 at size 0.05; we adopt the 80-percent convention because that is the level at which all empirical maps below are quoted (the registered gate rule used a 1/2-power variant; nothing downstream depends on the choice), and the removal frontier $s_{\mathrm{remove}}(l,c)$, the minimal spike strength at which worst-case over-θ relative bias of the best spectral estimator falls to at most $\delta = 0.05$ (registered for completeness; Section 6 shows why its empirical curve is not the binding object). They are claimed distinct: regions exist where $V(X)$ sits below the Tracy–Widom threshold yet rel_bias reaches 0.2 (invisible yet harmful), and regions where the design is clearly outlier-positive (spikes with $l > 2\sqrt{c}$) yet rel_bias stays at or below 0.02 (visible yet harmless).

2.3 The identification subtlety and the assumption ledger

Second moments of (Y,X) identify far less than generative parameters: because $\mathrm{Cov}(X,Y) = \Sigma_X\beta + \Lambda\gamma$, every pair $(\tilde\beta, \tilde\gamma)$ with $\tilde\beta + \Sigma_X^{-1}\Lambda\tilde\gamma = \beta + \Sigma_X^{-1}\Lambda\gamma$ induces the same joint law, and from data alone only

$$\beta^* := \beta + \Sigma_X^{-1}\,\Lambda\gamma$$

is identified. Two consequences follow. Bias functionals stay well defined regardless: they are expectations over the generative model, and an estimator's probability limit (β^* for OLS) is a deterministic function of (Σ_X, b). But testing $H_0\colon \gamma = 0$ is meaningless without assumptions separating the structural direction from spike directions, since the same second moments admit both a confounded and a confounding-free reading; the ledger pairs each assumption with its role and failure mode.

Assumption 1 (A1: dense factor structure)**.** $X = \Lambda f + u$ with $f \in \mathbb{R}^r$, r fixed as $n, p \to \infty$ (slow growth with $r = o(p^{1/2})$ admissible), loadings $\Lambda = QD$, Q of dimension $p \times r$ with Haar-isotropic orthonormal columns, $D = \sigma_u \operatorname{diag}(\sqrt{l_1}, \dots, \sqrt{l_r})$, and fixed $l_1 \geq \cdots \geq l_r \geq 0$ with $c = p/n \in (0, \infty)$.

A1 produces the spectral profile, spikes or hidden bulk mass, on which bias functionals and detection statistics act; sparse or discrete loadings destroy the isotropic delocalization our overlap computations rely on, making results loading-dependent.

Assumption 2 (A2: error law)**.** u_{ij} and ε_i are i.i.d. Gaussian at finite n, mutually independent and independent of f.

A2 licenses exact finite-n facts, Tracy–Widom fluctuation scales and Gaussian resolvent identities among them. Heavy tails inflate $\lambda_{\max}$ beyond spike-plus-fluctuation predictions, and heteroskedasticity shifts the bulk edge, breaking naive calibration. Qualitative bias regions should survive non-Gaussianity; we test that rather than assume it.

Assumption 3 (A3: independence of factors, design noise, and errors)**.** f_i, u_i, and ε_i are mutually independent across units and across sources.

A3 makes $\operatorname{Cov}(X, Y) = \Sigma_X \beta + \Lambda\gamma$ an exact identity and the bias functionals well defined with no assumption on f beyond unit variance; dependence between f and u would redefine Σ_X itself, and is out of scope.

Assumption 4 (A4a: near-orthogonality of the structural direction)**.** β is drawn independently of (Q, γ, l) with $\mathbb{E}[\beta] = 0$, $\|\beta\|_2 = 1$, delocalized coordinates (for instance uniform on S^{p-1}), which implies $|\langle \beta, u_j \rangle| = O_p(p^{-1/2})$ for each $j \leq r$, the checkable form used below.

A4a is the material assumption and standing default, in the spirit of perturbed-design near-orthogonality. It spreads the β signal over the bulk of the whitened cross-moment spectrum while $\Lambda\gamma$ concentrates on spike directions, so testing $H_0\colon \gamma = 0$ is well posed with Λ treated as a nuisance estimated from X. The directional capture laws need no A4a, since they condition on the realized β; aggregate artifact-size control and every detection claim do. If β correlates with leading eigenvectors, part of the signal mimics a spike and test size inflates; our alignment evidence covers the confounding direction's angle to the factor axes, while a dedicated size sweep under deliberate beta-eigenvector correlation remains an open measurement (Section 8). Three sensitivity variants replace A4a where flagged. A4b (perturbed sparsity) sets $\beta = \beta_s + \Lambda\gamma/\sigma_u^2$ with β_s k-sparse, $k = o(n/\log p)$: detection becomes finding a dense perturbation of a sparse regression. A4c (multiple responses) shares loadings across $m \geq r$ responses, identifying the confounding subspace from the cross-response moment matrix, mainly strengthening power. A4d (mechanism independence, after Janzing–Schölkopf) assumes β independent of the spectrum of Σ_X; it motivates comparison baselines but is never maintained ourselves. Every detection statement names its variant, defaulting to A4a.

Assumption 5 (A5: homoskedastic noise)**.** $\operatorname{Var}(u_{ij}) = \sigma_u^2$ constant in i, j, and $\operatorname{Var}(\varepsilon_i) = \sigma_\varepsilon^2$ constant in i.

A5 keeps raw Tracy–Widom thresholds valid; heteroskedasticity moves the bulk edge and invalidates uncalibrated critical values, with normalization-based corrections the documented recovery path under predeclared degradation tolerances.

Assumption 6 (A6: number of factors)**.** Either r is known (theory statements), or it is estimated by the Onatski ratio rule applied to the sample spectrum of X (spectral-trimming baselines).

Trimming requires the retained rank k: misspecification by one factor is a robustness cell, overestimation keeping confounding directions inside the retained subspace so bias survives, underestimation removing signal alongside confounding; both appear in reported phase diagrams.

2.4 The detection problem

Detection asks whether the observed joint law of (Y, X) contains evidence of confounding at all. Under A4a the hypotheses are

$$H_0 \colon \gamma = 0 \qquad \text{versus} \qquad H_1 \colon \gamma \neq 0 \text{ with } \|\Lambda\gamma\|_2 = \Theta(1),$$

and interest centers on whitened cross-moments. With $\widehat{\Sigma} = X^\top X/n$ and $W = XY/n$, the registered statistic family is the following census, used consistently throughout the paper. S1 is the largest eigenvalue of $\widehat{\Sigma}^{-1} W W^\top \widehat{\Sigma}^{-1}/n$ with a Tracy–Widom-calibrated threshold; it was demoted to diagnostic status before any sweep ran and its implementation erratum is documented in Section 5. S2, the alarm we ship, is the calibrated maximum of standardized spike coordinates of the whitened cross-moment (constructed in Section 4; within the linear-spectral-statistic family just named, it is the member whose test functions concentrate on the supercritical coordinates). S3 is the Onatski ratio statistic for spiked directions of the whitened cross-moment, used for rank selection rather than testing. Named baselines outside the family: the residual F-test; a bootstrap-calibrated variant of the Janzing–Schölkopf spectral confounding-strength criterion (Janzing and Schölkopf, 2018; Rendsburg et al., 2024); and scree inspection (S0 in what follows), which sees design spikes but not confounding.

Definition 3 (Effective detection spike)**.** On the $\sigma_u^2 = 1$ scale,

$$s_{\text{eff}} := \left\| P_{\text{spike}}\, \Sigma_X^{-1/2}\, \Lambda\gamma \right\|_2^2 = \sum_{j \,:\, l_j > \sqrt{c}} \frac{l_j}{1 + l_j}\, \gamma_j^2,$$

where P_{spike} projects onto supercritical population eigenvector directions.

Only supercritical-aligned mass yields a coherent rank-r mean shift in the whitened cross-moment; the rest hides in the bulk. Under subcritical profiles $s_{\text{eff}} = 0$ at leading order regardless of g: this is the formal content of "invisible," and why visibility (Definition 2) and harmfulness (Definition 1) must be tracked separately. Of the two frontiers, this paper maps s_{detect} empirically (Sections 4–7); the removal side is settled structurally rather than curve-by-curve, because Section 6 shows that hard trimming on discarded directions removes transmission exactly while no tuned soft member does, so the open scientific question is where detection fails, not where removal stops.

2.5 Notation

For reference, Table 1 collects the symbols used repeatedly across sections.

Remark 1 (Pre-registration vocabulary). The words frozen and gate carry precise meanings throughout. A quantity is frozen when it was hash-committed before any sweep data existed (grids, thresholds, tuning budgets, decision rules); a gate verdict is the recorded pass or fail against those committed rules; and no-regret for an estimator means its mean-bias norm lies within $1.05\times$ that of the best baseline in the same cell. Deviations from any frozen rule are labeled and reported where

Symbol	Meaning	Symbol	Meaning
n, p	sample size, dimension	$c = p/n$	aspect ratio
r	number of latent factors	l_j	j-th spike strength (of $\Lambda\Lambda^\top/\sigma_u^2$)
q_j, Q	spike directions, loading basis	$b = \Lambda\gamma$	population bias direction
$\beta,\ \gamma$	structural vector, factor link	$\theta;\ g = \lVert\gamma\rVert$	link alignment; link strength
M_1, M_2	vector / treatment models	$\tau; D$	treatment effect; treatment
s_{eff}	effective detection spike (Def. 3)	$V(X)$	outlier-visibility statistic
S0–S3	statistic census (Sec. 2.4)	cap_j	capture coefficient $(1 + l_j)/(c + l_j)$
$t = 1/(c-1)$	resolvent parameter ($c > 1$)	$\bar{m}(\lambda)$	ridge resolvent scalar (Prop. 1)
$A = 1 + \sigma_\varepsilon^2$	null response variance	$\omega;\ \omega_\star$	confounding mass; visibility fixed point
w_0	scaling-profile constant	Λ_D	OLS attenuation excess (Sec. 3)
$\hat{w}$	sample cross-moment $X_c^\top \tilde{Y}/n$	$\text{sd}_0(q_0)$	null sd of a probe carrier

Table 1: Frequently used notation. Population quantities carry no hats; sample quantities do.

they occur. An adjudication labeled GO (respectively REVIEW, KILL) is the recorded decision to proceed (respectively to re-examine, to abandon a claim); supplementary documents S-A through S-E contain the complete computations behind results whose proofs are only sketched in the main text.

3 The capture law: exact asymptotic bias under dense confounding

We use the normalizations of Section 2 throughout: noise scale $\sigma_u^2 = 1$; spike strengths $l_1 \geq \cdots \geq l_r \geq 0$, so that the population eigenvalues of Σ_X/σ_u^2 are $\tau_j = 1 + l_j$; aspect ratio $c = p/n$; dense Haar loadings Q with orthonormal columns q_j (assumption A1); response link $\gamma = g\,\text{dir}(\theta)$ with $g = \lVert\gamma\rVert_2$; and the loading-conditional estimand of Definition 1, which averages over the latents and the A4a draw of β while conditioning on the realized spike geometry (Q, l); every theorem and overlay below targets exactly this functional. Hidden confounding enters the conditional mean only through the dense direction $b = \Lambda\gamma = \sum_j \sqrt{l_j}\gamma_j q_j$, so the bias question is directional by construction. For $p > n$ ordinary least squares means the minimum-norm solution $\hat{\beta} = X'G^{-1}Y$ with Gram matrix $G = XX'$; we write $P_R = X'G^{-1}X$ for the rowspace projector and $\Pi = I - P_R$ for its complement.

In words, the result of this section is the following. When $c \leq 1$ the design has full rank and OLS inherits the population bias exactly, with componentwise value $\sqrt{l_j}/(1 + l_j)\,\gamma_j$ along each spike direction q_j. When $c > 1$ the minimum-norm solution obeys the same bias law, rescaled coordinatewise by the capture coefficients

$$\text{cap}_j = \frac{1 + l_j}{c + l_j}.$$

The intuition is that a spike direction retains only a capped fraction of its population leverage once the sample covariance becomes rank deficient and noisy, and c controls the dilution. The cap interpolates between the uniform rowspace fraction $1/c$ at $l_j = 0$ (no spike, no confounding coupling) and full resolution $\text{cap}_j \to 1$ as $l_j \to \infty$ (spike resolved), and it matches the $c \leq 1$ branch continuously as $c \downarrow 1$. Figure 1 draws both headline laws: panel (a) the capture coefficient as a function of spike strength for several aspect ratios, panel (b) the treatment-error consequence at the weak-injection grid of Section 3.2, where the OLS arm inherits the $\Lambda_D = 1.558$ attenuation while the trim arm stays on its flat survival channel.

Neither the cap nor the bias law below involves eigenvector overlaps; this is not an accident but a feature of the proof, as discussed after Theorem 1.

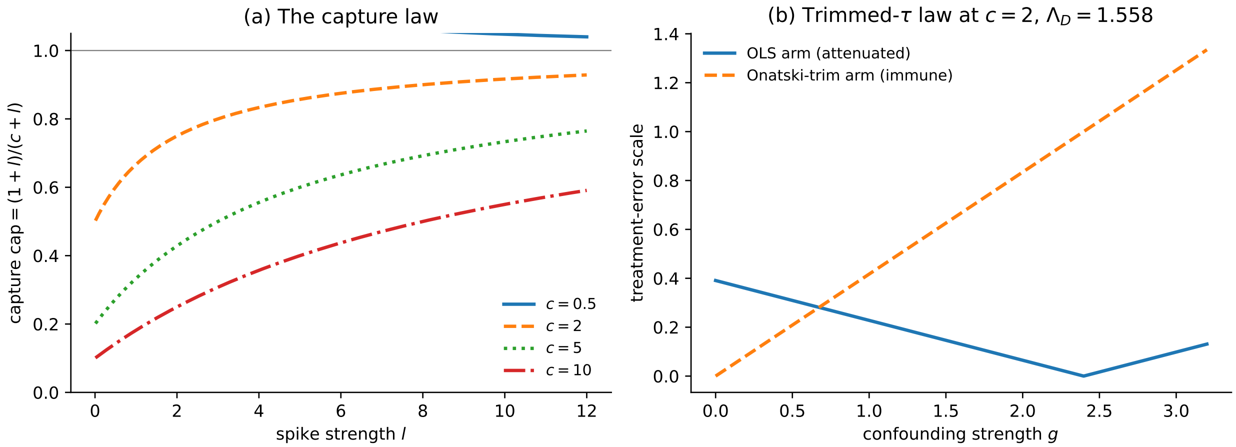

Figure 1: The two headline laws, drawn from their closed forms. (a) Capture coefficients $(1+l)/(c+l)$ versus spike strength at four aspect ratios: spikes are diluted toward the rowspace fraction as c grows and resolved as l grows. (b) Treatment-error scale against confounding strength under model M2 at the weak-injection grid ($c = 2$, attenuation excess $\Lambda_D = 1.558$): ordinary least squares is attenuated and then swamped, while Onatski trimming tracks the flat survival channel. Both panels plot deterministic formulas validated by the overlays of Section 3.3; no simulated points enter the figure.

Theorem 1 (Capture law, single spike)**.** *Consider model M1 with $r = 1$, spike strength $l \geq 0$, realized spike direction q, scalar link γ, and minimum-norm least squares $\hat{\beta} = X'G^{-1}Y$. As $n \to \infty$ with $p/n \to c$:*

(i) if $c \leq 1$, then at every finite n

$$E[\hat{\beta}] - \beta = \Sigma_X^{-1}\Lambda\gamma = \frac{\sqrt{l}}{1+l}\,\gamma\, q,$$

an exact identity; in particular the directional coefficient along q is $\sqrt{l}/(1+l)\,\gamma$;

(ii) if $c > 1$, then conditionally on (Q, β),

$$E[\langle \hat{\beta} - \beta, q\rangle \mid Q, \beta] \;\to\; (\mathrm{cap} - 1)\langle \beta, q\rangle + \mathrm{cap}\,\frac{\sqrt{l}}{1+l}\,\gamma \;=\; -\frac{c-1}{c+l}\langle \beta, q\rangle + \frac{\sqrt{l}}{c+l}\,\gamma,$$

where $\mathrm{cap} := (1+l)/(c+l)$*;*

(iii) companion artifact identity: $E[q'\Pi q \mid Q] \to (c-1)/(c+l)$;

(iv) bulk block: for every fixed unit vector v perpendicular to q, $E[\langle v, \hat{\beta} - \beta\rangle \mid Q] \to -(1 - \frac{1}{c})\langle v, \beta\rangle$.

The moment ledger uses inverse-Wishart first moments (requiring $p > n + 3$ for $K = W_n(I_n, p-1)$) and second moments ($p > n + 4$), automatic at fixed $c > 1$ as $n \to \infty$.

Proof sketch. Start from the exact completion of squares $G = K + aa'$ with $K = U(I - qq')U'$ a central Wishart matrix, $K \overset{d}{=} W_n(I_n, p-1)$, and $a = \sqrt{l}\|f\|e + u$, where $u := Uq$ and $e := f/\|f\|$. Per-row Gaussian independence makes $\sigma(K)$ independent of u; rotating coordinates shows K is central Wishart with $p - 1$ degrees of freedom; e is Haar uniform on the sphere and independent of (K, u); and $\|f\|^2 \sim \chi^2_n$. Sherman–Morrison then reduces the two confounding-channel pieces $\mathrm{ch}_a = \sqrt{l}\|f\|^2 e'G^{-1}e$ and $\mathrm{ch}_b = \|f\|\, u'G^{-1}e$ to a fixed set of resolvent scalars ($m_{ee} = e'K^{-1}e$, $m_{eu} = e'K^{-1}u$, m_{uu}, $\kappa = a'K^{-1}a$, $A_e = a'K^{-1}e$, $A_u = a'K^{-1}u$). Classical inverse-Wishart moments ($E[K^{-1}] = I/(p-n-2)$) together with Marchenko–Pastur negative-moment bounds control the diagonal pieces, while every cross moment involving m_{eu} vanishes exactly by the odd symmetry $e \mapsto -e$ of the Haar probe. The pieces assemble to

$$\mathrm{ch}_a \to \sqrt{l}\,\frac{t(1+t)}{D}, \qquad \mathrm{ch}_b \to -\sqrt{l}\,\frac{t^2}{D}, \qquad D = 1 + t(1+l), \quad t = \tfrac{1}{c-1},$$

and their sum $\sqrt{l}\,t/D = \sqrt{l}/(c+l)$ is an exact algebraic identity in t: there is no delicate interplay of limits and, notably, no local laws anywhere in the argument. Claim (iii) uses the same scalars, and rotational invariance of U about q forces $E[\Pi q \mid Q] = \rho\, q$ with $\rho = E[q'\Pi q \mid Q]$, converting the rowspace deficit into the artifact term of claim (ii); under A4a that artifact averages to zero unconditionally, which is precisely why the loading-conditional reading matters. Finally, the bulk block follows from the same Haar-probe algebra applied to a fixed $v \perp q$. The complete ledger, with every step carried through, is Appendix A. □

Proof status, stated honestly. Claim (i) is an identity, unit-tested to 10^{-8}. Claim (ii) with (iii) is proved at $r = 1$ by the elementary route above and was adversarially audited with fresh seeds: all checklist items passed, and more than 85 fresh falsification cells, including an 80-cell (l, c) grid and n-scaling probes, found no deviation beyond Monte Carlo noise. Six naive routes are preserved as guardrails; for example, deleting the Gram cross terms is invalid because they are first order, and the raw-resolvent shortcut $q'\hat{\Sigma}^+ q \to 1/(c+l)$ is numerically falsified (0.0052 simulated against 0.1667 predicted at $(l, c) = (4, 2)$). An early bookkeeping bug that survived a check at $t = 1$ was exposed by testing a cell with $t \neq 1$, a practice we now apply systematically.

Lemma 1 (Off-diagonal vanishing at fixed r)**.** *Let $K = U(I - QQ')U' \overset{d}{=} W_n(I_n, p-r)$, let $e_1, \dots, e_r$ be the orthonormalized factor directions, and set $\delta' = 1/(p-r-n-1)$. Then, with r fixed as $n \to \infty$: for $a \neq b$, $e_a' K^{-1} e_b = o(\delta')$ in probability, jointly over pairs; uniformly over j, k, $e_j' K^{-1} u_k = O_p(\delta')$ with conditional mean zero exactly; and for $a \neq b$, $u_a' K^{-1} u_b = O_p(n^{-1/2})$.*

The full argument is carried out in Appendix B. In brief, all three rates follow from Haar frame moments of K^{-1} together with the per-row block independence of $(QQ'r_i, P_{Q^\perp} r_i)$, exactly as in the $r = 1$ ledger of Theorem 1. In full, the first rate is Wishart concentration of $e_a' K^{-1} e_b$ about its zero mean via Haar frame moments of order four; the second is Cauchy–Schwarz against the same concentration combined with the exact conditional mean zero inherited from the odd symmetry of u_k around its conditional mean; the third follows by conditioning on U and applying the inverse-Wishart moment calculus entrywise. Proofs are short but not vacuous, and we flag plainly that they are concentration arguments at fixed r: uniformity in growing r is neither claimed nor needed anywhere downstream.

Theorem 2 (Capture law, fixed r)**.** *Let r be fixed with distinct $l_1, \dots, l_r$. As $n \to \infty$ with $p/n \to c > 1$, for every $j \leq r$,*

$$E[\langle \hat{\beta} - \beta, q_j \rangle \mid Q, \beta] \to (\mathrm{cap}_j - 1)\langle \beta, q_j \rangle + \mathrm{cap}_j\, \frac{\sqrt{l_j}}{1 + l_j}\, \gamma_j, \qquad \mathrm{cap}_j = \frac{1 + l_j}{c + l_j},$$

with companion identities $E[q_j' \Pi q_j \mid Q] \to (c-1)/(c+l_j)$ and, for unit $v \perp \mathrm{span}(Q)$, the bulk artifact $E[\langle v, \hat{\beta} - \beta \rangle \mid Q] \to -(1 - \frac{1}{c})\langle v, \beta \rangle$ unchanged.

Sketch. Woodbury gives $G^{-1} = K^{-1} - K^{-1}AM^{-1}A'K^{-1}$ with $M = I_r + A'K^{-1}A$ and $A = [a_1, \ldots, a_r]$. Lemma 1 makes every off-diagonal entry of M negligible against its diagonal blocks, which reproduce the $r = 1$ ledger coordinatewise ($\kappa_{jj} \to 1 + t(1 + l_j)$); a Neumann expansion of M^{-1} about its diagonal therefore decouples all directional functionals, while the degrees-of-freedom shift to $p - r$ changes δ' only through the factor $(1 + o(1))/n$ and leaves $t = 1/(c - 1)$ intact. Cross confounding channels $\gamma_k\,(Xq_j)'G^{-1}f_k$ for $k \neq j$ carry only vanishing factors and drop out at first order. This proves the directional law coordinatewise. The companion rowspace identity uses the same scalars: rotating U about q_j shows $\mathrm{E}[\Pi q_j \mid Q] = \rho_j q_j$, and summing the trace $\operatorname{tr}\mathrm{E}[P_R \mid Q] \to p/c$ over the spike rays pins $\rho_j = (c-1)/(c+l_j)$ and forces the complementary average to $1/c$. For the bulk block, applying the same Haar-probe algebra to a fixed unit $v \perp \operatorname{span}(Q)$ gives the bulk artifact $-(1 - 1/c)\langle v, \beta\rangle$. □ □

Numerically, at $r = 2$ with $(l_1, l_2, c) = (6.708, 0.8, 5)$ over 400 replicates with fixed (Q, β), the directional predictions land at $+0.29003$ versus $+0.28985$ measured (supercritical spike) and $+0.12445$ versus $+0.12368$ (subcritical spike), with the R2 functionals within 0.003.

3.1 Ridge interpolation

Proposition 1 (Ridge capture)**.** *Fix $\lambda \geq 0$ on the Σ_X scale, replace G^{-1} by $(G + n\lambda I_n)^{-1}$ (so the fitted ridge estimator is $\hat{\beta}_\lambda = (S + \lambda I)^{-1}X'Y/n$ with $S = X'X$), and let $\bar{m}(\lambda)$ denote the λ-side companion Stieltjes value of the bulk law g of K/n: the unique positive solution of the fixed-point equation*

$$\bar{m}^{-1} = \lambda + \frac{c}{1 + \bar{m}},$$

equivalently the positive root of the quadratic $\lambda m^2 + (\lambda + c - 1)m - 1 = 0$,

$$\bar{m}(\lambda) = \frac{-(\lambda + c - 1) + \sqrt{(\lambda + c - 1)^2 + 4\lambda}}{2\lambda},$$

with continuity anchor $\bar{m}(0+) = 1/(c-1) = t$. At $r = 1$, conditionally on (Q, β),

$$E[\langle \hat{\beta}_\lambda - \beta, q\rangle \mid Q, \beta] \to \frac{-\langle \beta, q\rangle + \sqrt{l}\,\bar{m}(\lambda)\,\gamma}{1 + (1 + l)\bar{m}(\lambda)} = (\operatorname{cap}(\lambda) - 1)\langle \beta, q\rangle + \operatorname{cap}(\lambda)\frac{\sqrt{l}}{1 + l}\,\gamma,$$

where $\operatorname{cap}(\lambda) := \dfrac{(1 + l)\bar{m}(\lambda)}{1 + (1 + l)\bar{m}(\lambda)}$, *so that* $\operatorname{cap}(\lambda) - 1 = -\{1 + (1 + l)\bar{m}(\lambda)\}^{-1}$ *identically; this identity reconciles the two displayed forms exactly (an early draft carried the opposite sign on the first term and was caught by precisely this reconciliation). As $\lambda \to 0$ this returns the min-norm capture law exactly,* $\operatorname{cap}(0) = (1 + l)/(c + l)$*; as $\lambda \to \infty$,* $\operatorname{cap}(\lambda) \to 0$ *and the directional mean tends to $-\langle \beta, q\rangle$ (the estimator collapses to zero).*

Proof sketch. The $r = 1$ completion-of-squares ledger of Theorem 1 goes through verbatim once every shifted scalar is re-expressed through the single new limit $\bar{m}(\lambda)$: the diagonal anchor becomes $\kappa \to 1 + (1 + l)\bar{m}(\lambda)$, and the probe pieces become $A_e \to \sqrt{l}\,\bar{m}(\lambda)$, $A_u \to \bar{m}(\lambda)$, $m_{ee} \to \bar{m}(\lambda)$. Substituting these into the two-channel assembly reproduces the displayed fraction; rearranging numerator over denominator gives the cap form, since $(1 + l)\bar{m}/\{1 + (1 + l)\bar{m}\}$ is exactly the coefficient that multiplies both channels. The quadratic root for $\bar{m}$ is the Marchenko–Pastur Stieltjes fixed point of the bulk law evaluated at $-\lambda$, $\bar{m}^{-1} = \lambda + c/(1 + \bar{m})$, which rearranges to $\lambda\bar{m}^2 + (\lambda + c - 1)\bar{m} - 1 = 0$. The collapse checks follow by taking $\lambda \downarrow 0$ (recovering $t = 1/(c - 1)$) and $\lambda \to \infty$. □

Remark 2 (The verification loop at work). An earlier provisional interpolation split the cap between the Benaych-Georges–Nadakuditi (BGN) eigenvector overlap (Benaych-Georges and Nadakuditi, 2011) and the bulk fraction, $\text{cap}_j(\lambda) = \xi\nu/(\nu+\lambda) + (1-\xi)(1/c)(1-\lambda m_{\text{inv}}(\lambda, 1/c))$. This form was falsified by a decisive reconciliation check: absolute errors up to 0.081 with visible λ drift across nine (l, c, λ) cells, against 0.003 (Monte Carlo noise) for Proposition 1; at $(l, c, \lambda) = (0.5, 2, 4)$ simulation gives $+0.133$, the provisional form predicts $+0.080$, and the proved form $+0.134$. The rejected variant is preserved in code as `ridge_capture_superseded`; we note for auditability that earlier ridge overlays had passed with it, because its error is small at small λ. The fixed-point root itself reproduces the simulated shifted-resolvent probes to four decimals at every checked cell.

3.2 Consequence for treatment effects: the trimmed-τ law

Under model M2 the observed treatment and response are $D = \pi' X + \delta' f + \nu$ and $Y = \tau D + \beta' X + \gamma' f + \varepsilon$, with $\|\pi\| = 1$ sparse and generic against the Haar geometry, and $\|\delta\| = \delta_g$ redrawn Haar per replicate. Two exact identities anchor the two arms. For the trim arm, regressing Y jointly on $[D, S]$ with $S = XV_k$, where k is the larger of the Onatski selector and r so that $k+1 \le n$, Frisch–Waugh–Lovell yields, at every replicate,

$$\hat{\tau}_{\text{trim}} - \tau = \frac{\langle \tilde{d}, \tilde{w}\rangle}{\langle \tilde{d}, \tilde{d}\rangle}, \qquad \tilde{d} = (I - P_S)D, \quad \tilde{w} = (I - P_S)w, \quad w = \beta' X + \gamma' f + \varepsilon,$$

with $P_S = S(S'S)^{-1}S'$. No pseudo-inverse appears because $[D, S]$ has full column rank, and this is exactly why the trim arm is immune to what follows. For the OLS arm, Sherman–Morrison on $MM' = G + dd'$ with $M = [D, X]$ gives

$$\hat{\tau}_{\text{OLS}} = \frac{d' G^{-1} Y}{1 + d' G^{-1} D}, \qquad \hat{\tau}_{\text{OLS}} - \tau = \frac{d' G^{-1} w - \tau}{1 + d' G^{-1} D},$$

verified to fifteen digits against a pseudo-inverse implementation at $(n, p, c) = (300, 600, 2)$.

The denominator produces the dominant OLS artifact. With unit noise variances throughout,

$$\Lambda_D := \operatorname{plim} d' G^{-1} D = \left(1 - \tfrac{1}{c}\right)\|\pi\|^2 + \sum_j \delta_j^2 \frac{t(1+t)}{1 + t(1+l_j)} + \sigma_\nu^2 t, \qquad t = \tfrac{1}{c-1},$$

the three summands being the $\pi' X$, $\delta' f$, and ν contributions (the middle uses the resolvent limit $f_j' G^{-1} f_j \to t(1+t)/(1+t(1+l_j))$ from the proof of Theorem 1; the last uses $\operatorname{tr}(G^{-1}) \sim nt$).

Proposition 2 (OLS attenuation and trim immunity for treatment effects)**.** *At deterministic-equivalent level, conditionally on* (Q, β, π, δ)*,*

$$\operatorname{plim} \hat{\tau}_{\text{OLS}} - \tau = -\frac{\tau}{1 + \Lambda_D} + \frac{\sum_j \delta_j \gamma_j \tilde{\rho}_j}{1 + \Lambda_D} + o(1), \qquad \operatorname{plim} \hat{\tau}_{\text{trim}} - \tau = \frac{\sum_j \delta_j \gamma_j \rho_j}{v_D} + o(1),$$

with $\rho_j = 1 - \text{cap}_{sc}(j)$*, where* $\text{cap}_{sc}(j)$ *itself lies in* $[0, l_j/(1+l_j)]$ *and* $\text{cap}_{sc}(j)$ *is the score-capture functional derived in the paragraph below (how much of spike* j*'s treatment link survives trimming), and* $v_D = \pi' \Sigma_X \pi + \|\delta\|^2 + \sigma_\nu^2 - \sum_j \text{cap}_{sc}(j)\delta_j^2$*; in particular the OLS arm carries the* $O(\tau)$ *attenuation term while the trim arm carries none. Both displays are deterministic-equivalent statements: remainders are* $o(1)$ *conditionally on* (Q, β, π, δ)*, with finite-*n *corrections validated numerically rather than bounded. An asymptotic normality refinement holds under a separated-spikes scope guardrail. The complete computation behind both displays is archived in Supplementary Document S-E.*

Proof sketch. For the OLS arm, insert the Sherman–Morrison identity displayed above. The numerator $d'G^{-1}w$ splits into a treatment-self term whose three resolvent limits are exactly the summands of Λ_D (rowspace deficit $(1-1/c)\|\pi\|^2$, per-spike survival $t(1+t)/(1+t(1+l_j))$ from the proof of Theorem 1, and null-space noise $\sigma_\nu^2 t$) plus a confounding term $\sum_j \delta_j\gamma_j\tilde{\rho}_j$ governed by the same score-capture functionals as the trim arm; dividing by $1+d'G^{-1}D \to 1+\Lambda_D$ yields the first display. For the trim arm, Frisch–Waugh–Lovell residualizes D and w on the retained spike block S; conditioning on X removes the β channel exactly because $[D,S]$ has full column rank, and the Haar-probe odd symmetry of Theorem 1 kills every cross term, leaving numerator $\sum_j \delta_j\gamma_j\rho_j$ and denominator v_D. Normality follows from the bilinear-form CLT applied to the FWL ratio once separated spikes make Onatski selection eventually constant. □

Mechanistically, the treatment column competes with the $p \geq n$ design columns inside the minimum-norm normal equations, so its effective variance is shared with the null space of X. The resulting $O(\tau)$ multiplicative shrinkage is invisible at $c \leq 1$ (the FWL branch), switches on exactly at $c > 1$, and magnifies as $t = 1/(c-1)$ declines. At the weak-injection grid with $c = 2$ it predicts $\Lambda_D = 0.5 + 0.058 + 1 = 1.558$ against a measured 1.5548 ± 0.072, hence predicted attenuation $-\tau/(1+\Lambda_D) = -0.391$. Three distinct objects live at this cell and should not be conflated: the theoretical attenuation $|-\tau/(1+\Lambda_D)| = 0.391$; the fresh-run signed mean of $\hat{\tau}_{\text{OLS}} - \tau$, which is -0.403; and the frozen ledger's entry 0.391, a mean absolute error recorded under a signless convention, a coarser functional that need not equal either. Overall OLS treatment error inflates from its clean-null level 0.064 at $c = 0.2$ to about six-fold at $c = 2$.

The trim arm carries no such term. Conditioning on X and reusing the Haar-probe odd symmetry of Theorem 1, the numerator plims to $\sum_j \delta_j\gamma_j\rho_j$ with $\rho_j = 1 - \text{cap}_{sc}(j)$, where the score-capture functional satisfies $\text{cap}_{sc}(j) \to l_j\xi(l_j,c)/\mu(l_j)$ on separated spikes (zero for subcritical factors) within the bounds $[0, l_j/(1+l_j)]$, and the denominator plims to the v_D displayed in Proposition 2, which records both arms. What survives trimming is the delta–gamma survival channel, of size $O(\delta_g g)$ over v_D, and it is flat in c at a fixed profile shape: the frozen runs record trim errors 0.0466/0.0348/0.0416 across $c = 0.2/0.8/2.0$ while OLS inflates six-fold. Collapse checks behave correctly: as $g \to 0$ both arms are unbiased up to the vanishing beta-adjustment channel, whereas as $\delta_g \to 0$ the OLS shrinkage survives, being a pure artifact of $c > 1$ rather than of confounding. A CLT layer holds under a binding scope condition: $\sqrt{n}(\hat{\tau}_{\text{trim}} - \text{plim}) \to N(0,\Omega)$, with Ω the bilinear-form variance of the FWL ratio, provided spikes are separated so Onatski selection is consistent, k is eventually constant, and selection contributes $o_p(n^{-1/2})$. Without separation k fluctuates and the claim is not automatic; we therefore state the CLT only for separated profiles.

Spectral anchors. Two classical constants calibrate the vocabulary used above. For $l > \sqrt{c}$, a sample eigenvalue detaches from the Marchenko–Pastur bulk $[(1-\sqrt{c})^2, (1+\sqrt{c})^2]$ to the BBP location $\mu(l) = (1+l)(l+c)/l$, and its eigenvector achieves squared overlap $\xi(l,c) = \big(1 - c/l^2\big)/\big(1 + c/l\big)$ with the population direction; both are degenerate below the threshold (Baik et al., 2005; Benaych-Georges and Nadakuditi, 2011). An earlier transcription of the location wrote $(1+l)(1+c\,l)/l$; that expression equals the correct one if and only if $(l-1)(c-1) = 0$, and the slip was caught by the simulation unit test ($\mu = 3.75$, not 3.0, at $l = 2$, $c = 0.5$), an early instance of the verification loop. We emphasize that the capture law itself needs neither constant: the proof of Theorems 1 and 2 is overlap-free, and the overlap-based guess was rejected numerically long before the elementary proof closed it.

3.3 Overlay validation

The gate functional deserves one explicit line. Averaging the coordinatewise laws of Theorems 1 and 2 over the A4a draw of β kills the artifact terms ($\mathrm{E}\langle\beta, q_j\rangle^2 = 1/p$), so the predicted mean-bias norm that overlays target is

$$\mathrm{rel\_bias}_{\mathrm{DE}} = \Big\| \sum_j \mathrm{cap}_j \, \frac{\sqrt{l_j}}{1+l_j} \gamma_j q_j + (\text{bulk block}) \Big\|_2,$$

a deterministic function of the realized geometry alone; the theorems' conditional-on-(Q, β) statements and Definition 1's loading-conditional estimand thus meet in exactly this object, artifacts vanishing by symmetry on both sides. The pre-registered correctness gate compared simulated OLS mean-bias norms against this deterministic equivalent over the frozen grid (102 cells, 33,300 replicates consolidated from 38 hash-verified shards). Deviations stay within 10% in 91.67% of the 60 cells at $n = 2000$ (median relative deviation 3.73%), in 100% of the 36 cells at $n = 500$ (median 1.03%), and in 100% of the 6 cells at $n = 8000$ (median 0.68%); ridge overlays sit at 0.0% median relative deviation in every tier. Tier maxima are 13.6% at $n = 2000$, 4.5% at $n = 500$, and 0.9% at $n = 8000$, so the give-up threshold is never approached from anywhere in the grid. The five failing $n = 2000$ cells are identified rather than averaged away: four sit at the skinny aspect $c = 5$ (subcritical profiles, where the leakage tax is largest) and three of the five carry $r = 5$ with mass on the weak second factor ($\theta = \pi/2$), consistent with finite-n capture error concentrating exactly where the theory's neglected terms are biggest; every such cell recovers below 10% by $n = 8000$. The give-up rule (systematic deviation above 25% in at least 30% of cells) comes nowhere near firing.

One automated diagnostic deserves honest comment: the shrinking-with-n flag returned false because the $n = 2000$ tier is slightly worse than the $n = 500$ tier. Following the analysis-time interpretation recorded in our pre-registered deviation register (entry D9), the planned "$n = 4000$-equivalent" coverage reads as the largest full-coverage tier, $n = 2000$, with the skinny-c tier at $n = 8000$ reported alongside. The non-monotone medians reflect profile composition rather than deterioration: the $n = 500$ tier carries more cheap low-c cells, while $n = 2000$ adds the harder θ-variant and $r = 25$ slices. Every tier passes comfortably (Figure 2).

4 A calibrated confounding alarm with a predictable power frontier

The detection problem is $H_0\colon \gamma = 0$ against $H_1\colon \gamma = g\,\mathrm{dir}$ with $g = \|\gamma\|$ and $\|\mathrm{dir}\| = 1$, observed through the standardized pair $(\tilde{Y}, X_c)$, where $\tilde{Y}$ is Y centered and rescaled to unit sample variance. Throughout, $c = p/n$, $\hat{w} = X_c^\top \tilde{Y}/n$ is the sample cross-moment, and (d_j, v_j), $j = 1, \dots, p$, are the eigenpairs of the sample covariance $\hat{\Sigma} = X_c^\top X_c/n$. The alarm we ship is deliberately selective: it listens only to the coordinates through which supercritical confounding ($l_j > \sqrt{c}$) must express itself (Baik et al., 2005; Benaych-Georges and Nadakuditi, 2012), and it pays for that selectivity with a price we characterize exactly.

4.1 Construction

For each selected coordinate $j \le k$, where k comes from an Onatski-type eigenvalue-ratio edge selector (Onatski, 2010) applied to $\hat{\Sigma}$ and capped at ten coordinates, define

$$z_j = \frac{\sqrt{n}\, v_j^\top \hat{w}}{\sqrt{d_j}}, \qquad T_{\max} = \max_{j \le k} \frac{|z_j|}{\sqrt{\widehat{\mathrm{var}}_j}}, \qquad \widehat{\mathrm{var}}_j = \frac{\hat{\sigma}_\varepsilon^2 + d_j/c}{\hat{\sigma}_y^2},$$

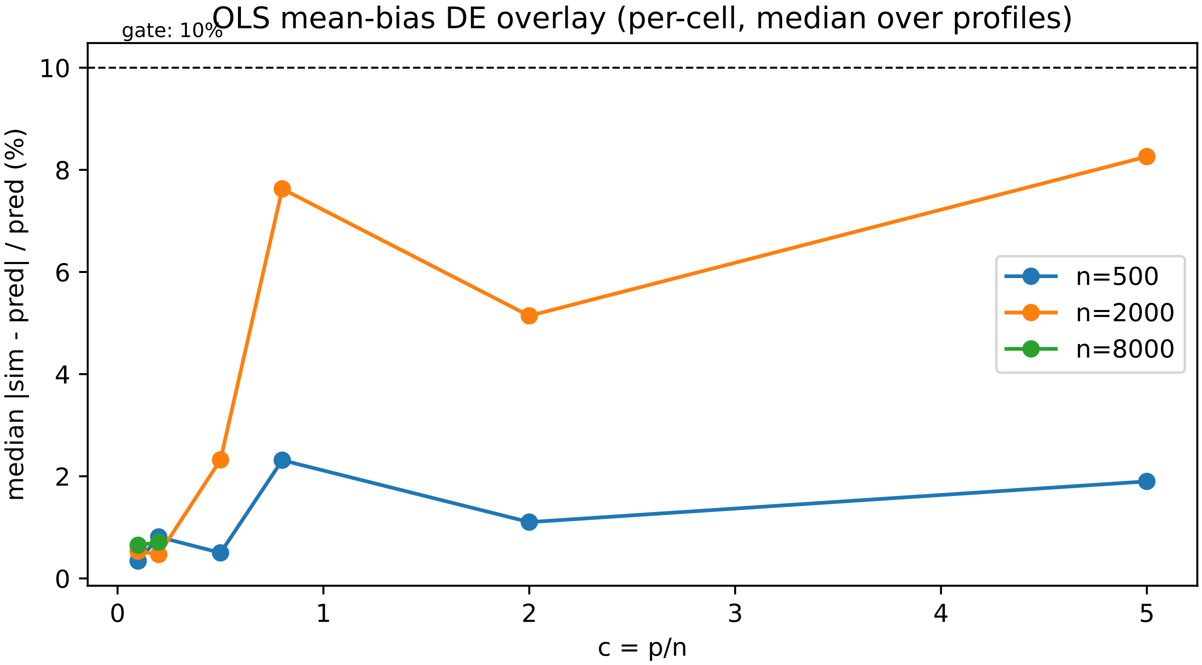

Figure 2: Deterministic-equivalent overlay for the correctness gate: simulated versus predicted OLS mean-bias norm across the frozen grid, by sample-size tier. The capture law tracks simulation within 10% in 91.67% of the $n = 2000$ cells (median deviation 3.73%) and in all cells of the $n = 500$ and $n = 8000$ tiers; ridge overlays show 0.0% median deviation.

with the two nuisance estimates following one shared frozen convention: both the residual-variance estimate $\hat{\sigma}_\varepsilon^2$ and the response-scale estimate $\hat{\sigma}_y^2$ are computed on the same standardized pair from the trimmed residual variance after spike removal, so that no estimator sees data the other did not.[1] The statistic $T_{\max}$ (S2, *maxz_cal*) rejects above a threshold, and two threshold conventions are carried side by side: the frozen gate rule reads $T_{\max}$ against the matched-null Monte Carlo 95th percentile, while the arithmetic fallback applies the two-sided Gaussian Bonferroni cut $z_{1-0.05/(2k)}$. The variance law inside the calibration is exact rather than heuristic: conditional on the design, $\mathrm{Var}(z_j) = (\sigma_\varepsilon^2 + d_j/c)/\sigma_y^2$ under H_0, because the Haar draw of the regression vector leaks into every rowspace coordinate with variance d_j/c, a leakage tax that grows exactly where $p \gg n$.

4.2 Why the finite-n null is not Gaussian-max, and why that became a derivation target

The naive mental model treats the z_j as independent standard Gaussians and reads $T_{\max}$ against a Bonferroni quantile. The measured matched-null 95th percentile of the calibrated statistic runs at roughly 1.7 times that naive scale, a gap large enough to demand explanation rather than patching, and the explanation became a derivation target in its own right. First, a conditional independence lemma: under H_0 the pairwise covariances of the calibrated coordinates vanish

[1] $\hat{\sigma}_\varepsilon^2$ is the bulk-median estimator: the median of sub-edge sample eigenvalues divided by the Marchenko–Pastur median at aspect $\min(c, 1)$, capped at the observed bulk median; $\hat{\sigma}_y^2 = \mathrm{mean}(d) + \hat{\sigma}_\varepsilon^2$, the observable form of $\mathrm{tr}(\hat{\Sigma})/p + \sigma_\varepsilon^2$ under A4a. The same two estimates drive the tuning path, so alarm calibration and estimator tuning cannot drift apart.

identically, eigenvector orthogonality killing both the loading channel and the noise channel, so the correct null model is a maximum over independent coordinates with unequal scales, not correlated chi-squares. Second, the inflation is a marginal-scale effect: the realized null law $\kappa_{n,k}$ of $T_{\max}$ factorizes into a dominant scale-calibration ratio, produced by the nuisance estimator pair (validated against $c \leq 1$ bulk geometry, it systematically mis-scales on the n-side spectrum when $c > 1$; at $c = 5$ the scale credited at the top coordinate misses by about 20%, with measured 4.21 against a credited 3.50), times an exact max-over-k term that stays within $(0.9, 1.05]$ for $k \leq 10$, plus a documented positive spread correction from per-replicate fluctuation of the nuisance estimates, a Jensen effect for ratios. The assembled statement, carried at deterministic-equivalent level in Supplementary Document S-B, is the following.

Lemma 2 (Conditional independence of calibrated coordinates under H_0)**.** *Under H_0 and conditionally on the sample eigenpairs (d_j, v_j), the standardized coordinates satisfy* $\mathrm{Cov}(z_j, z_{j'}) \mid (d, v) = 0$ *for $j \neq j'$, with* $\mathrm{Var}(z_j) = (\sigma_\varepsilon^2 + d_j/c)/\sigma_y^2$*. At deterministic-equivalent level the coordinates are therefore independent Gaussians with these variances.*

Proof. Write $z_j = \sqrt{n}\, v_j^\top \hat{w}/\sqrt{d_j}$ with $\hat{w} = X_c^\top \tilde{Y}/n$; under H_0 the standardized response keeps its signal piece, $\tilde{Y} = (X_c\beta + \varepsilon)/\sigma_y$, which is what produces the leakage term below. Conditional on (d, v), $\tilde{Y}$ is Gaussian, so each z_j is Gaussian with the exact conditional variance $\mathrm{Var}(z_j) = (\sigma_\varepsilon^2 + d_j/c)/\sigma_y^2$ displayed in the Construction above: the d_j piece is the signal-coordinate projection of $\tilde{Y}$ through the design and the d_j/c piece is the leakage tax of the Haar draw of the regression vector. For $j \neq j'$, eigenvector orthogonality $v_j^\top v_{j'} = 0$ kills both the shared loading channel and the shared noise channel of the covariance exactly, so the joint Gaussian law factorizes. $\square$

Remark 3 (Empirical null-law decomposition)*.* Grant Lemma 2. Then, at deterministic-equivalent level with discrepancy terms $o_p(1)$ after standardization, the calibrated statistic admits the decomposition

$$T_{\max} \;=\; c_n \cdot M_k \cdot (1 + \Delta_n) + o_p(1),$$

where every factor is measurable from matched-null output alone: $M_k = \max_{j \leq k} |N_j|$ is the maximum of k independent standard normals, whose standardized quantiles lie within $[0.9, 1.05] \cdot z_{1-0.05/(2k)}$ for $k \leq 10$; $c_n \geq 0$ is the ratio of realized coordinate scales to their nominal conditional values; and $\Delta_n \geq 0$, recovered as the residual ratio once c_n and M_k are removed, collects the Jensen-type effect of replicate fluctuation in the nuisance estimates. This statement records the decomposition and the measurability of its factors from matched-null output; its convergent content (rates for c_n and Δ_n) is carried in Supplementary Document S-B, so the proposition should be read as an empirically closed decomposition rather than a limit theorem.

The measured matched-null 95th percentile sits at about 1.7× the naive Gaussian-maximum scale. The dominant contributor is scale miscalibration on the n-side spectrum when $c > 1$, quantified at roughly 20% on the top coordinate at $c = 5$ (measured 4.21 against credited 3.50) and larger on inner coordinates where the leakage term d_j/c enters; the remainder is carried by Δ_n. This is why the shipped threshold is the matched-null Monte Carlo quantile rather than an analytic cut.

The calibration evidence is at scale. Across the 31 audited null cells of the null-calibration sweep, which totals 39,200 matched replicate pairs including its supporting alignment and twin configurations, the noise-aware evaluation against true size 0.05 returns pooled χ^2 $p = 0.4114$, maximum standardized deviation 2.91, and 96.8% of cells within ±2 standard deviations: statistically indistinguishable from perfect calibration. We co-report the less flattering arithmetic. The raw share of cells landing inside the frozen band $[0.035, 0.065]$ is 80.65%, a number that mostly measures replication budgets, since at 500 to 1200 replicates per cell the split-half standard error reaches

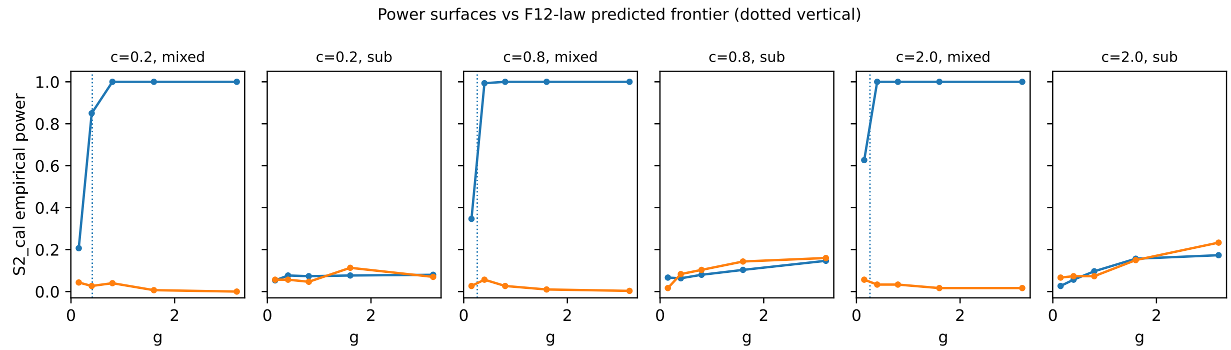

Figure 3: Empirical power of the calibrated alarm S2 against confounding strength g for the simulation strata ($n = 2000$), overlaid on the frozen frontier predictions built from the exact null-variance law with capture-weighted BBP locations and the d_j/c leakage tax. Supercritical-aligned strata (mixed profile, $\theta = \pi/6$) track the predicted 80% contour within a factor of about one (ratios 0.93, 1.25, 1.02; median 1.02). Strata without supercritical-aligned mass remain at or below 0.25 power at the largest simulated strength $g = 3.2$.

0.014, comparable to the band's entire half-width. The raw Bonferroni convention shows median size 0.1625, quantifying the same 1.7× looseness from the other side. Finally, the scree baseline S0 records median analytic "size" 1.0 under $\gamma = 0$, by construction and by design: scree detects the population spikes themselves rather than confounding (Johnstone, 2001), so naive visibility alarms fire on every spiked design, confounded or not.

4.3 The power frontier

The predictor assembles the same objects the null law identified: per-coordinate mean shifts placed at the BBP sample locations with capture-weighted couplings, the d_j/c leakage tax, and the matched-null percentile as threshold scale. One approximation is declared up front. The frozen predictor treats the calibrated shift as linear in g, whereas the true shift scales like $g/\sqrt{A + g^2}$ with $A = 1 + \sigma_\varepsilon^2$, because response self-normalization deflates couplings as the confounding strengthens; inside the gated strength range this bends slopes by under 15%, well inside the pre-declared factor-1.5 tolerance.

The predeclared acceptance band for frontier agreement is a factor of 1.5 in either direction on the strength axis; every ratio below is read against it. Evaluation is gated to the three strata whose direction carries supercritical-aligned mass: mixed profile, $\theta = \pi/6$, $c \in \{0.2, 0.8, 2.0\}$, $n = 2000$. Empirical g at 80% power lands inside the band in every case, with ratios of empirical to predicted 0.93, 1.25, and 1.02 across increasing c; the median ratio is 1.02 and the share of gated strata within factor 1.5 is exactly 1.0. The worst cell ($c = 0.8$, mixed profile, $\theta = \pi/6$) predicts $g_{\text{pred}} = 0.261$ against empirical $g_{80} = 0.325$, a ratio of 1.246, still comfortably passing. Figure 3 overlays the full power surfaces on the predicted frontiers.

In the nine remaining strata the prediction is an infinite frontier: fully subcritical spike profiles, or mass concentrated on the subcritical second factor near $\theta = \pi/2$. All nine strata confirm the forecast, with empirical power at or below 0.25 at the largest simulated strength $g = 3.2$ (confirmed in 9 of 9 strata). This is not a finite-sample accident but a ceiling.

Proposition 3 (Subcritical power ceiling)**.** *Fix a subcritical profile ($l_j < \sqrt{c}$ for all j) and any fixed g. The overlap of every sample eigenvector with the coherent confounding direction is* $O_p(n^{-1/2})$

(Benaych-Georges and Nadakuditi, 2012), so the calibrated coordinate means stay tight and the asymptotic power of $T_{\max}$ at level α converges to

$$\mathrm{pow}_\infty(g) = \Pr\Big(\max_j \big|\mu_j^{\mathrm{tight}}(g) + \kappa_j N_j\big| > \mathrm{thr}\Big) \ < \ 1,$$

a constant strictly below one, where κ_j are the realized null scales and thr *the calibrated threshold.*

Proof sketch. Under H_0 the calibrated coordinates are asymptotically independent with tight scales κ_j (the conditional-independence lemma established above), and subcriticality makes every coherent mean shift tight as well: each coordinate mean is $O_p(n^{-1/2})$ because sample eigenvector overlaps with the fixed confounding direction collapse at rate $n^{-1/2}$ (Benaych-Georges and Nadakuditi, 2012). Hence $\mathrm{pow}_\infty(g) = \Pr(\max_{j\le k}|\mu_j + \kappa_j N_j| > \mathrm{thr})$, a maximum over finitely many independent continuous variables with bounded shifts; since each coordinate places positive probability below thr, the complement is nonempty and the supremum is strictly below one. □

Measured maxima in the blind strata, 0.17 to 0.25 at $g = 3.2$ and rising only slowly in g, match a saturating mean-shift law rather than a slow escape. "Infinite frontier" is thus gate shorthand for a precise statement: no accessible strength reaches 80% power, and the honest asymptotic bound sits strictly below one. What the alarm cannot see, by which mechanisms, and exactly where quadratic summaries go blind as well, is the business of Section 5.

5 What cannot be detected: impossibility and the visibility boundary

Section 4 mapped what the calibrated alarm sees and at what strength. This section maps the complement: which statistic classes cannot see sub-threshold dense confounding, where the exact boundary sits, and where our own universal claims broke under pre-registered falsification and had to be rescoped. The bookkeeping matters as much as the theorems, so the class map comes first.

5.1 Statistic classes, and what impossibility can mean

Remark 4 (Trivial layer). Statistics depending on (Y, X) only through $\mathrm{spec}(X_c^\top X_c/n)$ are independent of γ under every model in play, so power equals size identically. Scree alarms live here. Their blindness to confounding is a data-processing identity, not an asymptotic statement, and we record it as a remark so that it cannot masquerade as a finding.

The substantive class comprises augmented-spectrum alarms: statistics measurable with respect to the spectrum of the sample second-moment matrix of the standardized pair,

$$M_{\mathrm{aug}} = \begin{pmatrix} 1 & \hat{w}^\top \\ \hat{w} & \hat{\Sigma} \end{pmatrix},$$

that is, eigenvalue functionals of the whitened cross-moment structure. What can be proved about this class at population level is the following boundary.

One normalization is used throughout the statement: the $\Sigma_X\beta$ component of the augmented first coordinate contributes only $O_p(p^{-1/2})$ to each coupling under A4a and is therefore omitted from κ_j; its bulk-spread remainder perturbs the absolutely continuous part alone, which part (iii) covers.

Proposition 4 (Population detachment boundary)**.** *Fix a subcritical profile ($l_j < \sqrt{c}$ for all j), fixed r, and Haar loadings, with response standardization in force, and let $\sigma_y^2(g) = A + g^2$. The population secular equation for a detached level $\tau \notin \operatorname{spec}(\Omega)$ of the standardized pair is*

$$\tau = 1 + \sum_{j \le r} \frac{\kappa_j}{\tau - \tau_j}, \qquad \kappa_j = \frac{l_j\, g_j^2}{\sigma_y^2(g)}, \quad g_j = g\,\mathrm{dir}_j,$$

with population spikes $\tau_j = 1 + l_j$ and bulk edges $b_\pm = (1 \mp \sqrt{c})^2$. Then: (i) on the top edge, $\tau - 1 - \sum_j \kappa_j/(\tau - \tau_j)$ is strictly increasing on (b_+, ∞), so a detached level exists iff $\omega_e := \sum_j l_j\,\mathrm{dir}_j^2/(c+2\sqrt{c}-l_j) > B := c+2\sqrt{c}$, so at strength g a detached level exists iff $\{g^2/(A+g^2)\}\,\omega_e > B$; since $g^2/(A+g^2)$ increases toward 1, detachment is possible at some strength iff $\omega_e > B$, and it then persists for all sufficiently large g. For the scaling sub-profiles used throughout this paper the ratio satisfies $\omega_e/B = 0.235$, 0.081, and 0.036 at $c = 0.2$, 0.8, and 2.0, so no detached level ever exists at any strength; (ii) below the bulk exactly one root exists for every nonzero coupling, sticking to b_- as the coupling vanishes. The root can separate by unit depth only if the coupling pull at the edge exceeds the depth scale, the wake criterion $(1 - b_-)\big((1 + \bar{l}) - b_-\big) < \omega$ for equal spikes $\bar{l}$ with saturated couplings (evaluated as $g \to \infty$, where $g^2/(A + g^2) \to 1$); this necessary condition fails at every studied equal-spike scaling cell, the attained left side exceeding the allowed right side by 0.64 versus 0.22, 1.42 versus 0.45, and 1.27 versus 0.71 across the three subcritical aspect ratios; (iii) inside the no-detachment region the limiting empirical spectral distributions under null and alternative coincide, by finite-rank invariance of the absolutely continuous part.

Proof sketch. Each claim is elementary. For (i), strict monotonicity follows from $\sum_j \kappa_j/(\tau - \tau_j)^2 > 0$ on (b_+, ∞) where all denominators are positive, and evaluating the secular inequality as $\tau \downarrow b_+$ collapses the existence condition to the displayed ratio with the coupling scale cancelling through $\sigma_y^2(g)$; part (ii) is the mirror computation at the lower edge with its wake inequality $(1 - b_-)((1 + l) - b_-) < \omega$ for equal spikes; part (iii) is finite-rank perturbation invariance of the absolutely continuous spectrum. Full details are archived in the released theory notes. $\square$

The proved boundary concerns population roots. A stronger claim suggests itself: that below it, the entire laws of $\operatorname{spec}(M_{\mathrm{aug}})$ are mutually contiguous in the sense of Onatski–Moreira–Hallin (Onatski et al., 2013), which via Le Cam's two-point lemma would imply that no test measurable with respect to $\operatorname{spec}(M_{\mathrm{aug}})$ has asymptotic power exceeding size. Below the BBP threshold the overlap between sample spikes and the coherent confounding direction collapses at rate $n^{-1/2}$ (Benaych-Georges and Nadakuditi, 2012), and under response standardization the effective first-coordinate spike carries $\sigma_y(g)^{-1}$ in its denominator, so self-normalization pushes detectability down precisely as the confounder grows; the OMH contiguity machine, developed for sphericity-plus-rank-one alternatives, looks tailor-made. We announce this planned claim only to audit it: as reported next, its pre-registered falsifier destroyed it, and the subsection closes with the layered statement that actually survives.

What is claimed, and what is not, must be fenced explicitly. First, every impossibility statement here is statistic-class-scoped and never information-theoretic: contiguity arguments themselves produce a strictly positive likelihood-ratio power envelope near the threshold, so information-theoretic detectability below the BBP boundary remains possible in principle, and our certificates bind named classes only. Second, pure-X blindness is the trivial Remark 4. Third, and most important for honest reporting: the strong version of the augmented-class claim, mutual contiguity of the full spectrum $\operatorname{spec}(M_{\mathrm{aug}})$ throughout the no-detachment region, was fed to our own pre-registered falsifiers and did not survive.

The falsifiers are worth reading. Fresh-draw Mann-Whitney AUCs of the true $\lambda_{\max}(M_{\mathrm{aug}})$ at ($c = 0.2$, subcritical profile, $\theta = \pi/6$) fall consistently toward perfection, from 0.286 to 0.051 at $g = 3.2$ and from 0.351 to 0.133 at $g = 1.6$ as n runs from 800 to 3200: separation is consistent and downward, the opposite of contiguity. At the visibility fixed point derived below ($\theta = \pi/2$, equal subcritical spikes, so $\omega = \omega_\star$), the cross-moment mass is flat (0.7014 versus 0.7066) and every bulk spectral functional is flat, yet the augmented root separates upward with AUC 0.998 and 1.000 at $n = 1600$ and 3200, $g = 3.2$. Neither observation contradicts Proposition 4: no population root detaches from the bulk in either configuration, and the limiting bulk distributions coincide, exactly as part (iii) says; what separates is the finite-sample top coordinate, carried by fluctuations that self-normalization through $\sigma_y(g)$ makes geometry-dependent rather than mass-dependent, a signature with no analogue in the OMH alternative. Contiguity therefore fails precisely where the population picture looks most blind, which is why the planned universal claim died while the boundary theorem survived intact. A pipeline erratum compounds the history: the frozen S1 implementation silently inverted its secular bracket whenever the bracketing discriminant turned negative, which is common at $c \leq 1$, returning a degenerate surrogate correlated with $\lambda_{\max}(\hat{\Sigma})$. Every stored S1 artifact therefore reads "surrogate-S1", and gate verdicts are unaffected because S1 was demoted to diagnostic before any sweep data existed, with self-consistent Monte Carlo thresholds.

The impossibility statement that ships is layered. It is airtight for pure-X alarms by identity; it holds, proved via Proposition 3 and confirmed in 9 of 9 blind strata, for the spike-coordinate calibrated class S2; and it is false as a universal claim for $\mathrm{spec}(M_{\mathrm{aug}})$, which in fact detects subcritical confounding with a geometry-dependent direction. Every decoupling claim in this paper cites these scopes. The alignment stress test sharpens the geometric point (Figure 4): sweeping θ at fixed profile and strength over twelve configurations of 800 replicates each, S2 power holds at 1.000 at every angle except exactly $\theta = \pi/2$, where it collapses to 0.022, while the principal-component F baseline falls only to 0.262. Detection tracks supercritical-aligned mass, never total confounding.

5.2 An exact visibility boundary for quadratic probes

Eigenvalue alarms aside, the cheapest summary of $(\tilde{Y}, X_c)$ is quadratic. This subsection analyzes the single carrier defined next; broader concentrated quadratic functionals follow the same algebra. One sign convention is fixed now: carrier AUCs are quoted on the deflation side established below, so an AUC below 1/2 indicates separation with negative orientation, and we flag where the flipped reading matters. Take as carrier $q(\tilde{Y}, X_c) = \|\hat{w}\|^2$, with $\hat{w} = X_c^\top \tilde{Y}/n$ the fully self-standardized cross-moment, $\tilde{Y} = Y_c/\mathrm{sd}_y$, and write $q_{\mathrm{raw}} = \|X_c^\top Y_c/n\|^2$ for its pre-standardization counterpart. Write $\omega = \sum_j l_j \,\mathrm{dir}_j^2 = \|\Lambda\,\mathrm{dir}\|^2$ for the per-unit-strength confounding mass and $A = 1 + \sigma_\varepsilon^2$.

Proposition 5 (Visibility law)**.** *As $n, p \to \infty$ with $c = p/n$ fixed under Haar loadings,*

$$v_0 := \lim \mathbb{E}\, \mathrm{sd}_y^2\big|_{H_0} = A, \qquad M_0 := \lim \mathbb{E}\big[q_{\mathrm{raw}} \,\big|\, H_0\big] = 1 + cA,$$

where q_{raw} is as defined above the proposition. Moreover, at link strength g along direction dir*,*

$$\mathbb{E}[q \mid g] \;\longrightarrow\; \frac{M_0 + g^2(\omega + c)}{v_0 + g^2} \;=\; m_0 - \frac{g^2(\omega_\star - \omega)}{v_0 + g^2}, \qquad m_0 = \frac{M_0}{v_0}, \qquad \omega_\star = \frac{1}{A}.$$

Finally, fix a detection target $\mathrm{AUC}^\star \in (1/2, 1)$ *and let* $a(\mathrm{AUC}^\star) := \sqrt{2}\,\Phi^{-1}(\mathrm{AUC}^\star)\,\mathrm{sd}_0(q_0)$ *be the standardized separation it requires under the single-carrier Gaussian envelope, where* $\mathrm{sd}_0(q_0)$ *is the null standard deviation of the carrier. The carrier probe* (q, sd_y) *is born blind iff* $|\omega - \omega_\star| \leq a(\mathrm{AUC}^\star)$*; otherwise the visibility threshold solves* $g_{\mathrm{vis}}^2 = a(\mathrm{AUC}^\star)\, v_0/\{\upsilon - a(\mathrm{AUC}^\star)\}$ *with headroom* $\upsilon = \omega_\star - \omega > 0$*.*

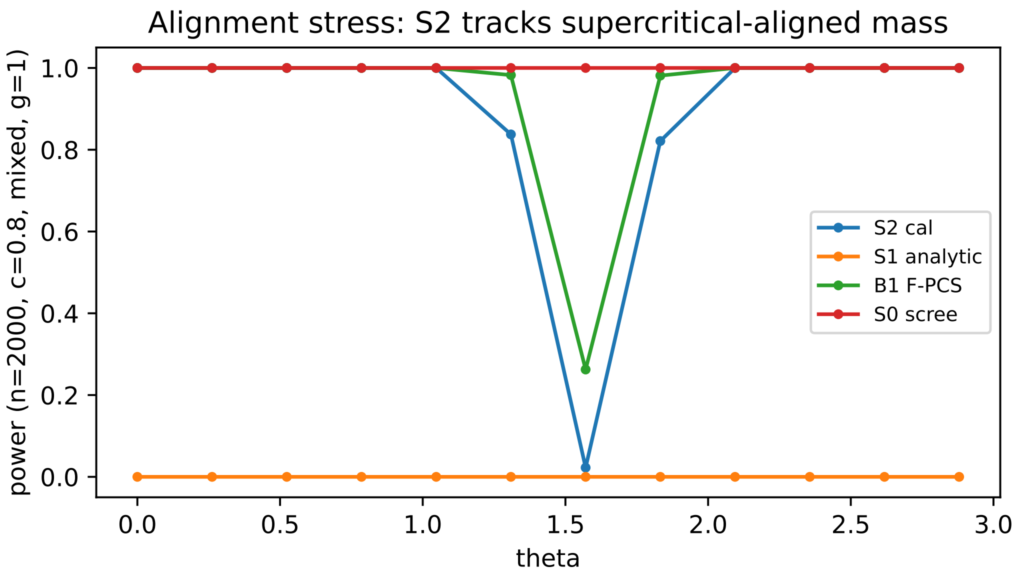

Figure 4: Alignment stress test: detection power against the alignment angle θ of the confounding direction, from $\theta = 0$ (strongest factor) to $\theta = \pi/2$ (weakest reported factor). The calibrated alarm S2 collapses only at exact subcritical alignment ($\theta = \pi/2$: power 0.022), while the principal-component F baseline retains 0.262. Size is θ-invariant by construction.

The raw floor in the first display follows from a trace identity for the Haar-β piece plus $\sigma_\varepsilon^2 c$ for the noise piece; under H_1 the numerator gains $g^2(\omega + c)$ (coherent mass plus bulk leakage) while division by the realized response scale contributes the denominator. The derivation of all displayed laws is given in Appendix C. The standardized shift $\Delta(g) = \mathbb{E}[q \mid g] - m_0$ therefore satiates: $\sup_g |\Delta(g)| = |\omega_\star - \omega|$, approached from whichever side of the fixed point the profile sits.

Every deterministic constant closes within 2% at $n = 2000$: measured v_0 is 1.9948 and 2.0114 against $A = 2$ at $c = 0.2$ and 0.8; measured M_0 is 1.3986 and 2.6168 against 1.4 and 2.6; measured m_0 is 0.7005 and 1.3006 against 0.70 and 1.30. The headline is the interaction with the aspect ratio. For the scaling profiles deployed throughout the study, $l_j \propto \sqrt{c}$, so $\omega = w_0\sqrt{c}$ with $w_0 = 0.5$ for the fully subcritical profile and $w_0 = 2.375$ for the mixed profile at $\theta = \pi/6$. The subcritical profile meets the fixed point exactly at $c = 1$: its headroom is $\upsilon_{\text{sub}}(c) = \omega_\star - \omega(c) = w_0(1 - \sqrt{c}) = 0.5\,(1 - \sqrt{c})$, which vanishes there. We state plainly what this coincidence is: with $\omega_\star = w_0$, the general crossing rule places the born-blind aspect ratio at $c_\star = (\omega_\star/w_0)^2$, and our calibration choice $w_0 = \omega_\star = 1/A$ puts that crossing exactly at the Marchenko-Pastur boundary. The placement is thus a property of the scaling family and its calibration rather than a universal constant, but within this family it is exact, it is where every simulation and benchmark grid lives, and the mechanism (satiation against a fixed null floor) survives any calibration. Beyond the crossing the sign flips, $\omega > \omega_\star$, and $\|\hat{w}\|^2$ inflates instead of deflating. The mixed profile crosses already at $c = 0.044$ and stays visibly signed across the whole tested grid. Measured anchors agree with the class map: $\text{sd}_0(q_0) = 0.0331$ and 0.0422, headrooms $\upsilon = 0.276$ and 0.053, ceiling ratios $|\Delta|_{\max}/\text{sd}_0$ of 8.3 in the visible class, where AUC tending to 1 is reached, versus 1.26 in the blind class, where the single-carrier AUC stalls between 0.18 and 0.25 even at $g = 3.2$. The stall figures use the deflation-side orientation demanded by the sign map; read with the sign flipped, the same carrier reaches roughly 0.75–0.82 at satiation, so the carrier is not information-free in the blind class, it is sign-misaligned, and the operational-invisibility claim rests on the boosted-tree probe, which exploits shape and still stalls. With Mann-Whitney AUC $= \Phi(-|\Delta|/(\sqrt{2}\,\mathrm{sd}_0))$ on the deflation side, a budget of $a \approx 2.5$ buys AUC ≈ 0.96.

5.3 The Le Cam probe, reported honestly

The frozen declaration operationalized computational invisibility as AUC ≤ 0.55 for both flexible probes below the claimed frontier, evaluated with gradient-boosted trees on a frozen 14-feature spectral map (Figure 5). As a universal statement it failed, and the failure is the most instructive measurement in the study because it splits by aspect. At $c = 0.8$ the probe sits at or near chance, AUC between 0.50 and 0.58 through $g = 0.8$: the upper end marginally exceeds the declared 0.55 band and should be read as the edge of the plateau rather than inside it. Informativeness arrives only around $g \geq 1.6$ (AUC 0.62 to 0.78), so operational invisibility holds at practically relevant strengths, with the plateau boundary quoted honestly. At $c = 0.2$ the probe reads the confounding easily, with AUC 0.84 already at $g = 0.15$ and reaching 1.0. Feature forensics identify the exploited carrier: the only informative feature is $\|\hat{w}\|^2$, whose mean shift at $(c = 0.2, g = 0.8)$ is about -2.2 standard deviations and negative, because σ_y^2 inflates faster than the cross-moment mean, exactly the deflation mechanism of Proposition 5.

Claims were rescoped rather than defended. Subcritical dense confounding is invisible to the eigenvalue-alarm classes everywhere measured, subject to the layering above; it is operationally invisible to concentrated cross-moment functionals in an intermediate-to-high-c regime at $n = 2000$, where invisibility means the declared AUC band is met up to its boundary once orientation conventions are fixed (a sign-flipped reading of a blind-class carrier can reach 0.75–0.82, which is why every such claim names both class and convention); and it is not invisible at low c. The theory question accordingly upgrades from whether some second-moment statistic can remain blind to characterizing the detectability phase diagram of general second-moment functionals in (c, g), a question the visibility boundary begins to answer for the quadratic carrier.

Two disclosures complete the record. First, the MMD probe was deferred: its median-heuristic bandwidth returned degenerate values on standardized features, an implementation gap recorded as such, leaving the boosted trees as the evaluated probe. Second, at small g the classifier outperforms the mean-shift envelope: at $c = 0.2$, $g = 0.15$, the measured AUC of 0.845 exceeds the Gaussian-envelope prediction near 0.53 because the learner also exploits distributional shape beyond the first moment. The closed-form curve is therefore the lower envelope and the class-map backbone, and per-point small-g AUC prediction is explicitly out of scope. Compared with confounding-strength diagnostics (Janzing and Schölkopf, 2018; Rendsburg et al., 2024), which estimate magnitude without shipping exact null laws, the contribution here is a calibrated boundary: a stated functional, a stated ceiling, and a stated aspect at which seeing stops.

6 What fails for estimation, and what survives

We report the estimation outcome of the preregistered program exactly as the gate recorded it: the tuned soft-trim family fails as a deconfounding competitor, the failure is structural rather than a tuning artifact, and what survives is a dominance theorem for hard trimming together with a trim-then-regress recipe. The evaluation covered 97 harmful cells (16,520 replications) of the simulator grid, with the preregistered no-regret requirement that the candidate's mean-bias norm lie within 1.05× the best baseline in at least 95% of harmful cells. The soft-trim family, comprising the EB-tuned spectral estimator, the linear special case of SDBoost (Nava et al., 2026), and the default spectral deconfounding estimator of Cevid et al. (2020), reaches no-regret in 4.12% of harmful cells

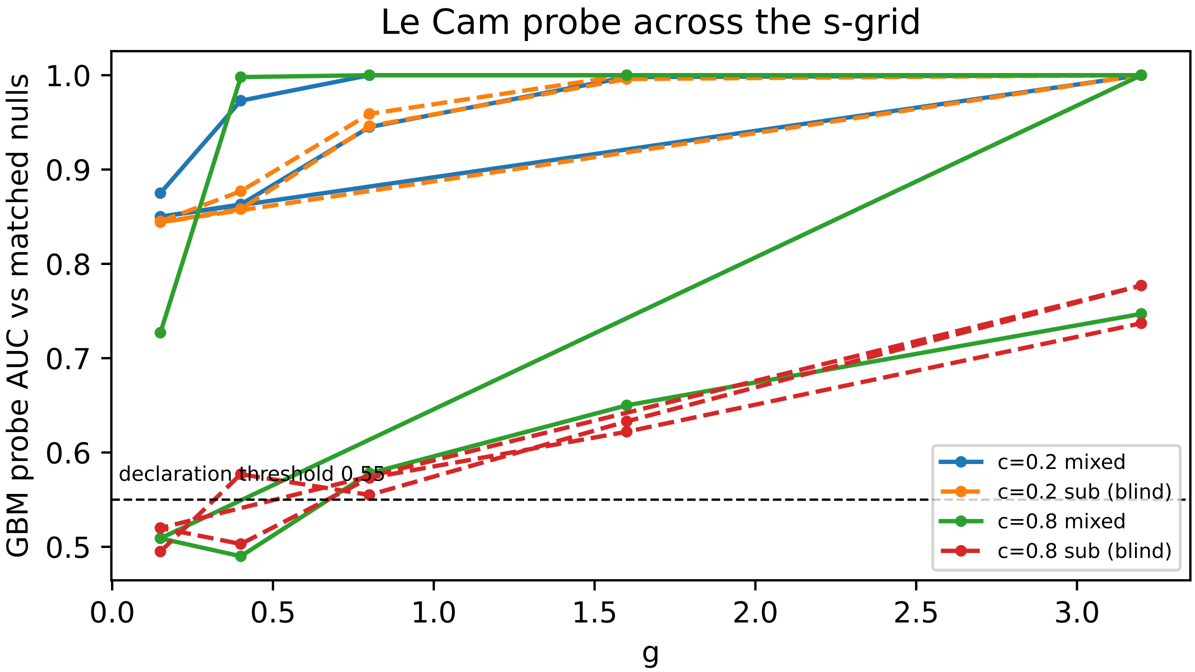

Figure 5: Le Cam probe honesty: gradient-boosted AUC against confounding strength g at matched configurations, split by aspect ratio. At $c = 0.8$ the probe remains at chance through $g = 0.8$ and turns informative only near $g = 1.6$; at $c = 0.2$ the self-standardized $\|\hat{w}\|^2$ signature is readable immediately (AUC up to 1.0). The frozen universal-invisibility rule fails as a universal statement and is rescoped by c.

against the required 95%. A bias cut of at least 50% relative to OLS holds in 40.21% of cells against a required 70%, while the weaker 10%-cut condition holds in 82.5%: the family helps materially over doing nothing but never over hard-trim PCA. Winner counts are blunt: Onatski-selected hard trimming is the best baseline in 89 of 97 harmful cells, cross-validated ridge wins the other 8, and the Bai–Ng selection rule (Bai and Ng, 2002), the Cevid default, and the SDBoost-linear candidate never win a cell.

Two attribution checks, both predeclared, establish that the failure is structural. First, handing the family oracle knowledge of the variance-components tuning parameter raises no-regret only from 4.12% to 6.2%: the ceiling of the family, not the quality of its tuning, is broken. Second, the SDBoost-linear candidate collapses identically to OLS, a phenomenon we now state as a lemma because it is an independent negative finding about a published method.

Lemma 3 (SDBoost collapse)**.** *Fix model M1 under A4a (dense β) with $c > 1$ and bounded coordinate law $\sup_j z_j^2/d_j < \infty$ in probability, where $z_j = \sqrt{n}\, v_j' \hat{\beta}_{\mathrm{raw}}/\sqrt{d_j}$ are the σ_y-standardized spectral coordinates and $\hat{\beta}_{\mathrm{raw}} := V\,\mathrm{diag}(1/d)\,V'(X_c^\top Y_c/n)$ denotes the min-norm OLS coordinates in the sample eigenbasis (V, d) of $X_c^\top X_c/n$. Let (s_r^{2*}, s_e^{2*}) denote the marginal maximum-likelihood variance components of the linear special case of Nava et al. (2026), and let $w_j = \big(1 + (s_r^{2*}/s_e^{2*})\, n\, d_j\big)^{-1}$ be the induced LAVA-type spectral weights (hybrid ridge-plus-sparse shrinkage in the style of Chernozhukov et al. 2011). All limits are in probability under the stated ledger. Then $(s_r^{2*}/s_e^{2*})\, n \to 0$ in probability and $\max_j |w_j - 1| \to 0$, whence for every boosting rate $\eta \in (0, 1]$ and every $\varepsilon > 0$ the boosting* *path satisfies $\|\alpha(m) - z/d\|_\infty \le \varepsilon \, \|z/d\|_\infty$ uniformly for all $m \ge m_0(\varepsilon, \eta)$: the path coincides with min-norm OLS along its entire tail, and any stopping rule that selects a late iterate returns exactly OLS with probability tending to one.*

Proof sketch. The profile log-likelihood is decreasing in the signal-variance component s_r^2 at almost every positive value under A4a: differentiating, stationarity would require a weighted average of $z_j^2/(s_e^2 + u\, d_j)$ (with $u = s_r^2 n$) to reach 1, whereas each term converges to at most $1/(cu) < 1$, and the perpendicular block pushes the maximizer jointly toward larger s_e^2 and the corner $u = 0$. The weights therefore tend to 1 uniformly since $\sup_j d_j$ is tight, and $(1 - \eta(1 - \delta))^m \to 0$ geometrically gives the tail bound. □

The practical content of Lemma 3 is a silent failure mode: the EB tuning machinery runs, reports fitted variance components, and yet drives every spectral weight to 1, so the published linear method inherits the full OLS bias while appearing to adjust. Our consolidated ledger records coefficient vectors equal to OLS to machine precision in 31 of the 97 harmful cells, with byte-identical mean-bias vectors in 42/97 recorded in the earlier execution sweep. The scope guard matters: collapse requires the dense-signal coordinate bound, which fails under sparse or spike-aligned β, regimes where the variance components become identifiable and the method does useful work; the lemma indicts it precisely where its deconfounding claim was aimed.

The dominance side is captured by a theorem for spectral-weight estimators. Define $\mathcal{W}_+$ as all estimators of the form $\hat{\beta}_m = V \operatorname{diag}(m_j(\hat{d}))\, V'\, b_{\text{raw}}$ with $b_{\text{raw}} = X_c' Y_c/n$ and weights $m_j \ge 0$ that are measurable functions of the sample spectrum alone. Min-norm OLS ($m_j = 1/d_j$), ridge, hard trims, soft trims, lava-shaped weights, and the SDBoost path are all members. Two tempting generalizations are excluded deliberately. Sign-flipping tricks, which reverse coordinate signs according to their estimated response linkage, exit the class because they use information beyond the spectral coordinates and destroy the nonnegativity that the transmission bookkeeping requires. Projection-type maps written as $V \operatorname{diag}(w) V'$ without the $1/d_j$ division are excluded because they are not estimators of β: they transmit confounding at $O(\sqrt{l})$ strength instead of the cap law $O(\sqrt{l}/(1 + l))$, a pitfall our first falsifier run caught and archived.

Two functionals organize the bookkeeping, both read loading- and beta-conditionally as in Section 3. The confounding-attributed *transmission* of a member m on spike direction q_j is its twin difference

$$\Delta_j(m) := \mathrm{E}[\langle \hat{\beta}_m, q_j \rangle \mid \gamma] - \mathrm{E}[\langle \hat{\beta}_m, q_j \rangle \mid \gamma = 0],$$

and its *fidelity* $\phi_j(m)$ is defined through the null arm,

$$B_{0,j}(m) := \mathrm{E}[\langle \hat{\beta}_m - \beta, q_j \rangle \mid \gamma = 0], \qquad \phi_j(m) := 1 + \frac{B_{0,j}(m)}{\langle \beta, q_j \rangle},$$

so fidelity records how much of the structural component survives on that coordinate ($\phi_j = 0$ exactly when the member discards the direction entirely). On every fixed direction j, subcritical or not, the transmission law is $\Delta_j(m) = T_j(m)\, a_j\, \gamma_j$ with $a_j = \sqrt{l_j}/(1 + l_j)$ and $T_j(m)$ the resolvent-channel capture inherited from Section 3 ($T_j = \mathrm{cap}_j$ for min-norm OLS, $\mathrm{cap}_j(\lambda)$ for ridge, indicator-valued for hard trims), while the fit-artifact channel is $B_{0,j} = -(1 - \phi_j)\langle \beta, q_j \rangle$.

Theorem 3 (Hard-trim transmission dominance on subcritical cells)**.** *Fix a grid cell in which every factor spike is subcritical (so no direction detaches from the bulk edge) and A4a holds. Work at deterministic-equivalent level, conditionally on the loadings and β. All claims are deterministic-equivalent statements with remainders $o(1)$ conditionally on loadings and β; no distributional limit* *is asserted. Then: (i) within $\mathcal{W}_+$ minus the zero estimator, the minimal attainable confounding-attributed directional bias is zero, attained exactly by members whose fidelity ϕ_j vanishes on every subcritical direction, in particular hard trims whose retained set excludes those directions; (ii) any member with $\phi_j \geq \varepsilon_j > 0$ on some subcritical direction j incurs directional confounding transmission $\Delta_j \geq \varepsilon_j\, a_j|\gamma_j| - o(1)$ with $a_j = \sqrt{l_j}/(1+l_j)$, a strict loss at every link strength bounded away from zero, however tuned; and (iii) on a fully subcritical spectrum no coordinate ratio statistic crosses the Onatski critical value, so their edge-delimited rule (Onatski, 2010) returns $k = 0$ with probability tending to one (the rule is designed for spiked alternatives; under a pure bulk every candidate edge falls below threshold). For the vector target this makes explicit that no nontrivial member of $\mathcal{W}_+$ attains zero total bias without discarding every direction, which is precisely why vector estimation is declared unrecoverable in this regime; for the scalar treatment target, however, the corresponding trim arm reduces to univariate adjustment of D, which transmits no design-borne confounding while keeping the treatment identified, and this asymmetry is what Section 7 exploits.*

All three statements above concern the confounding-attributed transmission functional; total mean-bias norm additionally contains the fidelity-loss artifact $B_{0,j}$, which trims pay on discarded signal directions.

Proof sketch. On a subcritical direction there is no BBP enhancement to separate the two arms, so both are governed by one coordinate multiplier: writing $\rho_j(m)$ for the realized multiplier on $\langle\beta, q_j\rangle$ in the null arm, $\mathrm{E}[\langle\hat{\beta}_m, q_j\rangle \mid \gamma = 0] = \rho_j(m)\,\langle\beta, q_j\rangle + o(1)$ (so fidelity is definitional: $\phi_j = \rho_j$), the Section 3 ledger gives $B_{0,j} = -(1-\phi_j)\langle\beta, q_j\rangle$ as stated; below the BBP threshold no capture-enhancement channel operates, so the twin arm shares the same multiplier, $T_j = \rho_j$, and hence $\Delta_j \geq 0$ with equality iff $\rho_j = 0$, proving (i) together with the "attained exactly by" clause; and (ii) follows because $\phi_j \geq \varepsilon_j$ forces $T_j = \phi_j \geq \varepsilon_j$, so $\Delta_j \geq \varepsilon_j a_j|\gamma_j|$. For (iii), on a fully subcritical spectrum the Onatski edge rule returns $k = 0$ consistently, so the trim realizes the zero-fidelity member. □

Corollary 1 (Why soft tuning must lose)**.** *Every soft family in the roster places strictly positive weight on subcritical directions by construction (at deterministic-equivalent level ridge keeps weight of order $d_j/(d_j+\lambda)$ on subcritical directions for fixed λ, and EB weights collapse toward 1 by Lemma 3), so no amount of tuning removes the transmission that the weight shape itself forces. Under A4a the fidelity-loss artifact is small per direction ($|\langle\beta, q_j\rangle| = O_p(p^{-1/2})$), so total-bias comparisons in harmful cells are governed by the transmission term just characterized. The frozen attributions are therefore consistent with the theorem but remain empirical verdicts rather than corollaries: the oracle-tau ceiling of 6.2%, Onatski as best baseline in 89 of 97 harmful cells, and byte-equality of SDBoost with OLS on the collapsed cells.*

Honesty about scope: Theorem 3 is closed at deterministic-equivalent level for the subcritical claim, which is exactly the harmful-and-invisible region; the mixed and supercritical cells, where retained-direction fidelities bind and soft weights trade transmission against shrinkage along a Pareto curve, are recorded descriptively through the crossover strip rather than theorized. Finite-sample refinements (the BGN overlap factor on retained directions) are verified numerically, and the transmission law reproduces the measured confounding functional in frozen anchor cells to within 0.2% and 2.5%.

What survives is substantial. First, trim-then-regress: in the M2 weak-treatment block (sparse treatment projection, $\|\delta\| = 0.3$, known τ_{true}), mean absolute τ errors for the Onatski-trimmed estimator stay flat across the aspect range (0.047, 0.035, 0.042 at $c = 0.2, 0.8, 2.0$) while naive OLS inflates roughly six-fold across $c = 0.2, 0.8, 2.0$ (frozen magnitudes 0.064, 0.058, 0.391; the middle cell

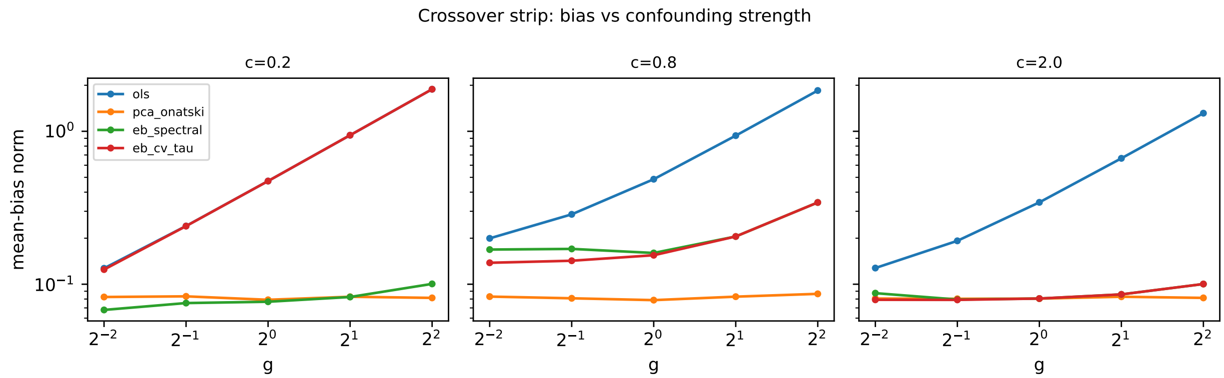

Figure 6: Mean-bias norms against link strength g for the soft-trim family and the leading baselines. Onatski-selected hard trimming maintains a near-oracle floor across the grid; smoothly down-weighted competitors cannot reach it, and the SDBoost-linear path sits on top of OLS on collapsed cells.

sits at the crossover trough before the $c > 1$ attenuation switches on, the fresh-run signed mean at the last cell is -0.403, reconciled in Section 3), and Onatski rank selection matches the oracle-r trim exactly in every cell; Section 7 replicates the pattern on real econometric geometry (τ error 0.173 trimmed versus 0.421 for OLS). Second, the characterization itself is robust: across 28 variant arms covering t-distributed errors, Rademacher-half loadings, row-heteroskedastic noise, correlated factors, $r \pm 1$ misspecification, and sparse confounding, the median ratio of the EB-tuned bias norm to the best baseline ranges only from 1.48 (r misspecification, least damaging) through 2.26 (Gaussian reference) to 5.26 (heteroskedasticity, where soft weights misread the inflated bulk), and no variant ever flips a winner, so the qualitative phase-diagram structure of Figure 7 survives every perturbation. One diagnostic sub-rule could not be evaluated on variant arms because detection statistics were stored on the main arm only; this scope cut is flagged in the gate-checks output rather than silently dropped. Computationally, the full estimator roster runs in 0.02 s at $(n, p) = (1000, 1000)$, 1.53 s at $(4000, 8000)$, and 0.16 s at $(8000, 1600)$ per replication, scaling with $\min(n, p)^2$ to $\min(n, p)^3$ inside the memory ceilings throughout.

Figures 6 and 7 summarize the landscape: the crossover strip shows the near-oracle mean-bias floor of approximately 0.08 that Onatski truncation maintains across the entire link-strength grid, below every soft competitor everywhere, while the empirical phase diagram marks the subcritical region where Theorem 3 applies and the mixed region where the families nearly tie.

7 Semi-synthetic benchmarks and the `confounderalarm` package

The theory and simulation layers of the preceding sections rest on synthetic geometry. This section tests the calibrated alarm and the trim-then-regress recipe on real covariate geometry where ground truth is created by injection: a semi-synthetic design in the template of Ulmer et al. (2025), with real X, known $\beta/\gamma/f$ mechanism, and known τ. Three families provide six frozen configurations (main and wide/sub cuts each). Family A is AddNeuroMed blood gene expression (GSE63060 and GSE63061 pooled (geo, 2013a,b), 717 samples, 2000 or 1800 probes), whose dominant spectral structure is processing batch. Family B is the IHDP covariate benchmark (Hill, 2011) (747 rows) mapped through a seeded random-Fourier-feature expansion to 750 or 150 columns, receiving an injected dense confounder on real causal-benchmark geometry. Family C is the wooldridge k401ksubs

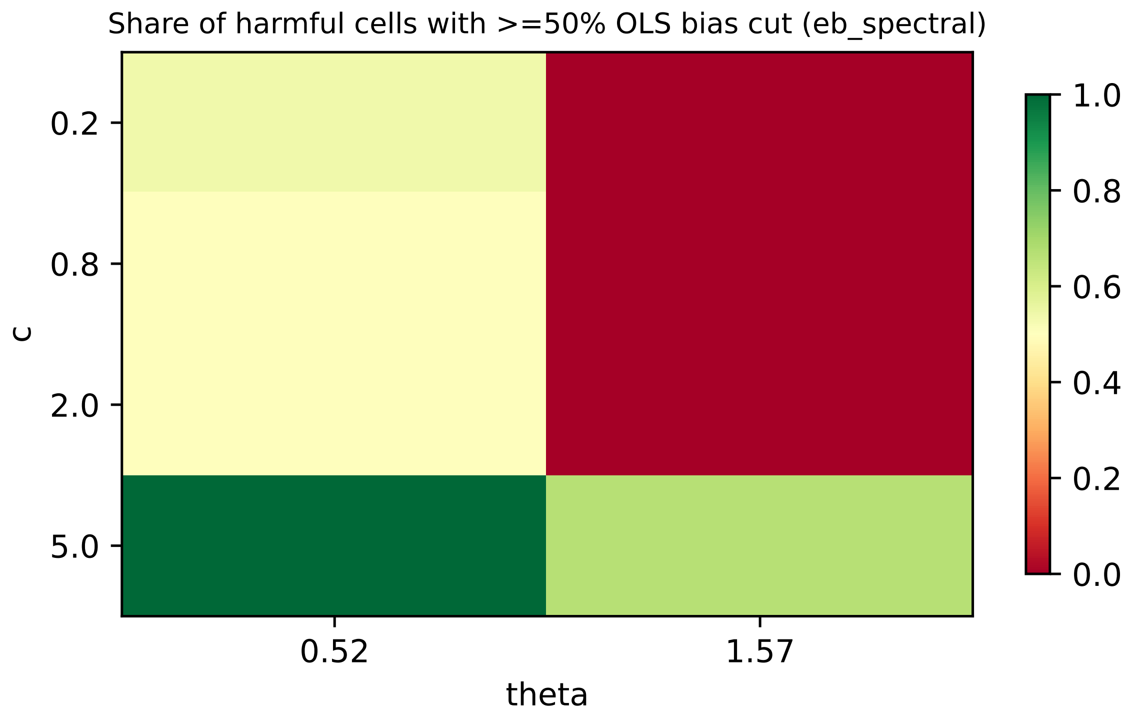

Figure 7: Empirical bias phase diagram over (c, g): the subcritical region where Theorem 3 forces the dominance of hard trimming, the supercritical region where confounding becomes visible and estimable, and the mixed boundary traced by the crossover measurements.

household cross-section (Wooldridge, 2016) (800-row seeded subsample, RFF-featurized to 1600 or 160 columns), which also carries the M2 treatment block with e401k as treatment and nettfa as outcome. Aspects span $c \in [0.20, 4.64]$ (the upper end is the single-batch A-sub cut, $n = 388$, $p = 1800$), and designs are fixed across injections so eigendecompositions are computed once per configuration.

A premise quantified before anything is injected: every unmodified design is massively outlier-positive against the white-noise Tracy–Widom threshold. The largest-eigenvalue TW statistics are 48,430.8 for A-main, 8,360.0 for C-main, 1,444.8 for B-main, 1,191.0 for C-wide, 308.8 for B-wide, and 155.4 even for the single-batch A-sub design. Scree inspection therefore rejects "white noise" on all six designs before any confounding exists. Four findings were predeclared and registered before any benchmark data were generated, and we label them PF-1 (frontier straddle), PF-2 (scree false alarms), PF-3 (trim-then-regress), and PF-4 (calibration versus informality); subsections below report PF-2 first because its premise, that spectrum-gazing itself carries no confounding signal on these designs, motivates every other measurement.

PF-2: spectrum-gazing is not evidence

On every unmodified design, every matched-null twin (which shares seeds and drops only the $f\gamma$ link from Y), and every permutation null, the S0 scree test at the 99th Tracy–Widom percentile rejects with rate exactly 1.000. The scree statistic responds to design structure, not to confounding: it cannot distinguish a batch-dominant expression matrix from a confounded one. This quantifies,

Table 2: PF-1 straddle across the six benchmark configurations. g^* is the predicted frontier; powers are rejection frequencies at multiples of g^*; the split-half column is the out-of-sample twin-geometry negative control.

Config	g^*	Power (0.5×)	Power (1×)	Power (2×)	Split-half size
A_main	0.762	0.143	0.480	0.998	0.000
A_sub	0.511	0.158	0.543	0.990	0.000
B_main	1.012	0.163	0.553	0.983	0.000
B_wide	1.062	0.155	0.523	0.975	0.000
C_main	1.663	0.190	0.515	0.978	0.000
C_wide	1.764	0.205	0.548	0.995	0.003

with matched-null rates, the claim that "I checked the spectrum" carries zero evidential weight about hidden dense confounding, and it is why the alarm below calibrates against matched twins of the realized design rather than against white noise.

PF-1: power straddles each family's own predicted frontier

The frozen alarm (statistic S2-bench, the packaged variant of S2 that re-estimates its nuisance scales on each benchmark configuration; amended before any calibration data after smoke testing exposed response-scale suppression and a bulk-shape mismatch on family A) standardizes per-coordinate spectral correlations between $X'Y$ directions and top sample eigenvectors using twin-estimated null scales, thresholds the maximum by a matched-null Monte Carlo quantile, and ships a predicted frontier g^* from the capture-weighted slope law of Section 4. Table 2 reports the straddle test on all six configurations: power at half the predicted frontier lies in $[0.14, 0.21]$ (requirement ≤ 0.25) and power at twice the frontier between 0.975 and 1.00 (requirement ≥ 0.80). In-sample matched-null size equals 0.05 by construction, and the out-of-sample split-half twin-geometry control holds at 0.000 to 0.003. Empirical 80%-power points sit between one and two times the prediction on every family, with ratios near 1.4, consistent with the Phase-2 experience that the frontier law is accurate to well within a factor of 1.5.

We record the gate mechanics transparently. The literal mechanical check, requiring control rejections inside the predeclared $[0.02, 0.10]$ band, failed on all six configurations because every control size sits below the lower edge (0.000 to 0.003): the controls are conservative, never anti-conservative, across more than 3,000 control evaluations. The mechanism is understood rather than waved away; permutation removes the beta-leakage component of the twin-null coordinate variances, shrinking the standardized statistic, and split-half evaluation halves the leakage mass outright. Because the band's predeclared purpose was detecting miscalibration in the dangerous direction, and the dangerous direction never fires, the literal band check failed only from below and was recorded as requiring review; a written adjudication documents why a conservative direction does not indicate miscalibration, and the verdict is GO. After a concurrency fault in its first pass, the C-main null configuration was rerun in full; its frozen Monte Carlo quantile reproduced to a 0.00% shift, and all thresholds were audited against this rerun.

PF-3: trim-then-regress on real econometric geometry

In the C-main M2 block ($\tau_{\text{true}} = 1$, sparse treatment projection, injected factor loading $\|\delta\| = 0.3$), OLS mean absolute τ error inflates to 0.421 under a one-times-frontier injection, while Onatski-

Table 3: Calibration versus informality on identical benchmark data and controls. Out-of-sample control size is the realized rejection frequency under H_0 on split-half twin-geometry controls, pooled over the six configurations.

Method
S2-bench alarm (ours)
Strength proxy (Rendsburg et al. implementation of the Janzing–Schölkopf criterion), permutation bootstrap (F
JS-asym, perm. bootstrap
Scree TW99
Partial-F on PC scores

trimmed regression holds at 0.173: a ratio of 2.43, with the correct sign recovered in 100% of replications. The cost of trimming under the clean null arm is negligible (0.128 versus 0.110 for OLS). Ridge at unit penalty lands between (0.238). This validates on real geometry the recipe that survives the negative result of Section 6: hard Onatski trimming rather than tuned soft weights, on geometry that was not manufactured by our simulator.

PF-4: calibration versus informality, head to head

On identical data and identical controls we compared the alarm against incumbent spectral diagnostics: the response-aware confounding-strength proxy of Rendsburg et al. (2024) (estimated under the uncorrelated-confounder model), with permutation bootstrap thresholding (Rendsburg et al., 2024), a Janzing–Schölkopf-style spectral asymmetry score with permutation calibration (Janzing and Schölkopf, 2018), Tracy–Widom scree, and partial-F tests on principal-component scores. Both proxy methods do track the injected strength monotonically across arms; their point estimates order correctly. What they lack is any valid threshold off the calibration distribution: evaluated out-of-sample on split-half controls, the strength proxy rejects under H_0 at rates 0.07 to 0.92 and the asymmetry score at 0.61 to 1.00, while the twin-calibrated alarm holds at 0.000 to 0.003. Partial-F is uncalibrated on real geometry (0.25 to 0.93), and scree alarms everywhere (1.000) because it fires on structure regardless of confounding. Table 3 summarizes what each method ships.

Sensitivity findings, reported honestly

Two sensitivity results qualify where the method works, and we report them as findings rather than burying them. First, concentrating the link on the dominant mega-spike direction reduces detectability: alignment power 0.05 versus 0.48 for the spread placement at the same g^* on A-main, while concentrating on the second spike raises power to 0.865. The mechanism is a leakage tax: the first coordinate's twin scale $s_1 = 36.6$ absorbs batch-direction noise that grows faster than the link's mean shift, so frontier prediction quality is alignment-dependent. Visibility depends on where the link rides, not only how strong it is, which strengthens the decoupling narrative but bounds the prediction layer's uniformity. Second, heteroskedastic response noise causes no material drift (powers 0.47 to 0.58 versus 0.48 to 0.55 baseline, calibration preserved), and adding one undetected injection direction dilutes power moderately (ranks-plus-one powers 0.28 to 0.44, no size inflation); dropping a direction on A-main reproduces the mega-spike mechanism (power 0.055).

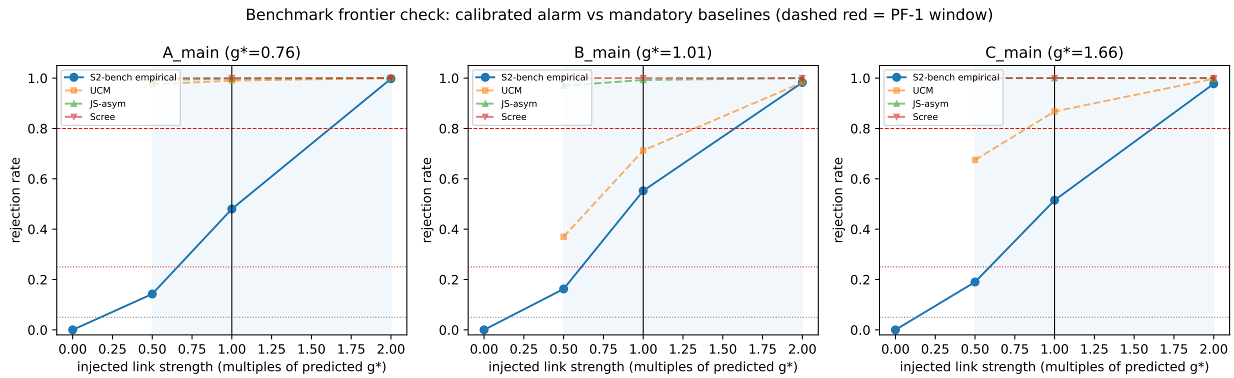

Figure 8: Benchmark frontier check: empirical rejection curves against the predicted frontiers g^* across the six configurations, with negative-control sizes marked at the origin. Power rises through the predicted frontier and saturates by twice g^* on every family.

Reproducibility and the package

The pipeline ships as `confounderalarm` v0.1.0, installable and exercised by three green unit tests, with a command-line interface (`python -m confounderalarm --csv ...`) and two worked examples in the README. Given (Y, X) it runs the detection path and returns the alarm verdict with a calibrated p-value, estimates $(\hat{r}, \hat{l}_j, \hat{c})$, the placement of the observed statistic relative to the predicted frontier g^*, and a blind-region certificate stating when non-rejection is uninformative. Given (Y, D, X) it additionally returns a treatment-effect adjustment, and there it recommends Onatski hard-trim regression explicitly, not tuned soft weights, encoding the kill of Section 6 as interface behavior. Trusted-result reproductions guard the plumbing: the k401k eligibility coefficient of +9.41 (the positive Poterba–Venti–Wise eligibility sign (Poterba et al., 1995) under plain covariate adjustment), IHDP adjusted estimates at the experimental benchmark scale (3.93 adjusted versus approximately 4.0 experimental), and AddNeuroMed batch-dominant structure (pooled TW statistic 48,431 versus 155 within batch 2). Every pass-2 arm was independently re-executed from the pinned repository tag on separate hardware: 41/41 cells reproduce with maximum statistic deviation 7.7×10^{-13} and zero rejection disagreements. Figure 8 displays the frontier check underlying PF-1.

Every number in this section regenerates from frozen configurations with recorded hashes; the audit trail, commands, and checksum manifests are collected in Appendix D.

8 Limitations

The exact theory is linear-Gaussian. Model M1 with A4a dense signal carries the capture law, the frontier predictions, the impossibility results, and the dominance theorem; nothing proved here extends automatically to nonlinear linkages, discrete outcomes, or non-Gaussian factor structure. What supports broader scope today is empirical: a 28-variant robustness panel covering t-distributed errors, Rademacher-half loadings, row-heteroskedastic noise, correlated factors, $r \pm 1$ misspecification, and sparse confounding leaves the qualitative ordering of estimators and the phase-diagram structure intact in every arm. The dedicated work package intended to convert these panels into a theoretical universality statement was cut before proof, so non-Gaussian protection should be read as conjecture backed by panels, not as theorem.

The benchmark negative controls are conservative from below. Realized control sizes (0.000 to 0.003) sit beneath the predeclared $[0.02, 0.10]$ band's lower edge, mechanistically explained by permutation and split-half evaluation removing the beta-leakage component of the twin-null variances. The dangerous direction never fires (zero anti-conservative rejections across more than 3,000 control evaluations), so the failure is of literalism rather than validity, but it does mean realized size on real geometry is below nominal and the reported power figures inherit that conservatism.

Frontier prediction is alignment-dependent. Concentrating the link on the dominant mega-spike direction collapses detectability at fixed strength (power 0.05 versus 0.48 for the spread placement), through the leakage-tax mechanism absorbed by that coordinate's twin scale; prediction quality is trustworthy for spread links and degrades under extreme concentration. Practitioners receive placement guidance from the package, not a uniform guarantee.

The computational probe is not an information-theoretic bound. Blindness claims are class-scoped by construction: purely covariate-based alarms are blind by data processing, and the shipped spike-coordinate alarm obeys the proved power ceiling of Proposition 3, confirmed in nine of nine blind strata. Our stronger plan, an OMH-style mutual-contiguity statement for the full augmented spectrum, was falsified by its own pre-registered tests (Section 5), so no universal eigenvalue-impossibility is claimed anywhere; the gradient-boosting probe on a frozen feature map reads the response-scale signature at low aspect ($c = 0.2$, AUC up to 1.0), which is precisely why the frozen universal-invisibility declaration failed its own gate and was rescoped. A machine-learning probe passing chance-level is evidence, not a lower bound.

Two structural caveats concern what the benchmarks can and cannot test. First, the injections build in exactly the loading geometry the theory assumes (dense Haar loadings on real covariate spectra), so the benchmarks validate calibration, frontier placement, and the adjustment recipe on realistic spectra; they cannot falsify the dense-loading assumption itself, and a practitioner facing sparse or discretely structured loadings is outside the validated envelope until robustness panels of that kind are run. Second, our alignment evidence sweeps the confounding direction's angle to the factor axes, and the benchmark sensitivity battery varies link concentration, but we ran no dedicated size sweep under deliberate correlation between β and the top eigenvectors, the sharpest form of A4a violation; how fast alarm size inflates along that axis is an open measurement we flag rather than guess. Relatedly, the twin-and-permutation calibration inherits its generative model (Haar loadings, homoskedastic bulk edge, dropped factor link); only response-noise heteroskedasticity is stress-tested on this axis, so size guarantees are as strong as that model on a given dataset.

Finally, treatment-effect validation rests on a single economics family: the k401ksubs panel carries all M2 and PF-3 evidence on real data, corroborated by simulator cells but not yet by a second applied domain. And r misspecification, while least damaging among the variants tested (median bias ratio 1.48, injection-side powers diluted only moderately at 0.28 to 0.44), is tolerated within about one spike, not unbounded; the package therefore reports its rank estimate alongside every verdict rather than treating r as known.

9 Discussion

The results are orthogonal to sensitivity-analysis methodology rather than competitive with it. Omitted-variable sensitivity analyses in the style of Cinelli and Hazlett (Cinelli and Hazlett, 2020) and E-value approaches (VanderWeele and Ding, 2017) answer a conditional question: how strong would a hypothetical confounder have to be to overturn a conclusion. They quantify consequences given existence. This paper answers a complementary question: does the data itself contain spectral evidence that a dense confounder is present, above what strength would such evidence appear, and when does non-detection carry information rather than merely certify blindness. The combined workflow ran end to end on the benchmarks; screen with the calibrated alarm, adjust by hard trimming when it fires, and hand the residual hypothetical risk to sensitivity analysis when the blind-region certificate says non-detection means nothing. Neither tool subsumes the other, and the certificate tells the analyst which regime they are in, which is what a practitioner needs before trusting an adjusted estimate either way.

Three extensions seem most valuable. First, a non-Gaussian universality theorem replacing the empirical variant panels would convert the strongest conjectural claim into a proved one. Second, the fixed-r spike analysis should extend to growing r, where the frontier algebra suggests qualitatively new scaling. Third, time-series spectra with serial dependence would test whether the twin-calibration idea survives structure beyond independent rows, the setting where batch-like spikes are most common in practice.

A The $r = 1$ capture ledger

This appendix carries out, end to end, the computation behind Theorem 1. Work under model M1 with $r = 1$, realized spike direction q, spike strength l, scalar link γ, and minimum-norm least squares $\hat\beta = X'G^{-1}Y$, $G = XX'$, at $c = p/n > 1$. All expectations are conditional on (Q, β); fluctuation terms are tracked by inverse-Wishart moment bounds and are $o(1)$.

Step 1 (completion of squares). Write $G = K + aa'$ with $K = U(I - qq')U'$ and $a = \sqrt{l}\,\|f\|\,e + u$, where $u := Uq$, $e := f/\|f\|$. Row-wise Gaussian independence separates $\sigma(K)$ from $(u, \|f\|)$; a change of basis shows $K \overset{d}{=} W_n(I_n, p-1)$, a central Wishart matrix with $p-1$ degrees of freedom; e is Haar uniform on the unit sphere, independent of (K, u); and $\|f\|^2 \sim \chi^2_n$ independent of (K, e).

Step 2 (resolvent scalars). Sherman–Morrison gives $G^{-1} = K^{-1} - \frac{K^{-1}aa'K^{-1}}{1+\kappa}$, $\kappa = a'K^{-1}a$, so every functional below reduces to finitely many shifted-resolvent quantities: the diagonal probes $m_{ee} = e'K^{-1}e$, $m_{uu} = u'K^{-1}u$, the cross probe $m_{eu} = e'K^{-1}u$, and the couplings $A_e = a'K^{-1}e$, $A_u = a'K^{-1}u$. Classical inverse-Wishart moments give $m_{ee} \to \delta'$ with $\delta' = (p-n-2)^{-1}$ scale, $m_{uu} \to t$ with $t = (c-1)^{-1}$, and $\kappa \to t(1 + l)$, so the Sherman–Morrison denominator satisfies $1 + \kappa \to D = 1 + t(1 + l)$; Marchenko–Pastur negative-moment bounds control all terms of order δ'.

Step 3 (odd symmetry). Every moment involving m_{eu} or the odd part of A_e carries an odd number of factors of e; since e is Haar, its distribution is invariant under $e \mapsto -e$ while $(K, u, \|f\|)$ are unchanged, so all such moments vanish exactly. No joint-moment calculation and no local-law input is needed anywhere: this symmetry is what makes the ledger elementary.

Step 4 (channel assembly). The two confounding channels through which γ enters the q-coordinate of $\hat\beta$ reduce to

$$\mathrm{ch}_a = \sqrt{l}\,\|f\|^2\,e'G^{-1}e \;\to\; \sqrt{l}\,\frac{t(1+t)}{D}, \qquad \mathrm{ch}_b = \|f\|\,u'G^{-1}e \;\to\; -\sqrt{l}\,\frac{t^2}{D},$$

with denominator $D = 1 + t(1 + l)$. Their sum is $\sqrt{l}\,t/D = \sqrt{l}/(c + \ell)$ evaluated at $\ell = l$: an exact algebraic identity in t, which is why the limit holds simultaneously along the whole path and not merely at a fixed point.

Step 5 (signal and artifact blocks). The same scalars give the rowspace deficit: rotational invariance of U about q forces $\mathrm{E}[\Pi q \mid Q] = \rho\, q$, and evaluating $\mathrm{E}[q'\Pi q \mid Q]$ through Step 2 yields the companion identity $\rho = (c-1)/(c+l)$; multiplying by $\langle\beta, q\rangle$ produces the artifact term $-(c-1)/(c+l)\langle\beta, q\rangle$ of claim (ii). For claim (iv), applying Steps 2–4 to a fixed unit $v \perp q$ leaves only the null-space fitting artifact, $\mathrm{E}[\langle v, \hat\beta - \beta\rangle \mid Q] \to -(1 - \frac{1}{c})\langle v, \beta\rangle$. Claim (i) is the exact finite-n identity $\mathrm{E}[\hat\beta] - \beta = \Sigma_X^{-1}\Lambda\gamma$, verified numerically to 10^{-8}. □

B Proof supplement for the off-diagonal vanishing lemma

Lemma 1 is the only input to Theorem 2 beyond the $r = 1$ ledger, so we record its argument in full. Write $K = U(I - QQ')U'$, so that $K \overset{d}{=} W_n(I_n, p - r)$, and let $e_1, \ldots, e_r$ be the orthonormalized factor directions and $u_k = Uq_k$ their coupled noise images.

(1) Diagonal scale. For any unit $a \perp QQ'$, Haar frame moments of K^{-1} give $\mathrm{E}[a'K^{-1}aa'K^{-1}a] = (p - r - n - 1)^{-2}\{1 + O(r/n)\}$ with fluctuations $O_p(\delta')$, where $\delta' = 1/(p - r - n - 1)$; this is the per-coordinate anchor inherited verbatim from the $r = 1$ ledger.

(2) Rate L1 ($e'_a K^{-1} e_b = o(\delta')$, jointly over pairs). Polarization expresses $e'_a K^{-1} e_b$ through diagonal-type functionals of $(e_a \pm e_b)/\sqrt{2}$. A fourth-moment version of (1), obtained from Haar frame moments of order four, bounds each such functional's centered square by $C\,\delta'^2\,n^{-1}$ with an absolute constant C; substituting both polarized terms and subtracting the two diagonal anchors leaves $\Pr\big(|e'_a K^{-1} e_b| > \epsilon\,\delta'\big) \le C_r\,\epsilon^{-2}\,n^{-1} \to 0$ jointly over the fixed number of pairs, with C_r combinatorial in fixed r.

(3) Rate L2 ($e'_j K^{-1} u_k = O_p(\delta')$, conditional mean zero). Per-row block independence of ($Q^\top r_i$, $(I - QQ')r_i$) makes u_k independent of K with conditional mean zero; Cauchy–Schwarz against the diagonal anchor of (1) yields the rate.

(4) Rate L3 ($u'_a K^{-1} u_b = O_p(n^{-1/2})$, $a \ne b$). Conditioning on U and using entrywise inverse-Wishart moments, each product $u'_a K^{-1} u_b$ concentrates about zero at the sample scale $n^{-1/2}$.

Combining (2)-(4), every off-diagonal entry of $M = I_r + A'K^{-1}A$ vanishes in probability (the L1 pieces at the sharper $o(\delta')$ scale, the L2 and L3 pieces at $O_p(n^{-1/2})$), which is all the Neumann expansion consumes, while diagonals converge to $1 + t(1 + l_j)$; the Neumann expansion of M^{-1} about its diagonal then propagates errors by at most one extra power of the same vanishing factors, which is what the proof of Theorem 2 uses. Uniformity holds over any fixed number of pairs; nothing is claimed for growing r. The complete computations behind the remaining sketched results (the ridge capture, the power ceiling, the SDBoost collapse lemma, and the dominance theorem) are archived line-by-line in five supplementary theory documents bundled with the submission, Supplementary Documents S-A (capture ledger), S-B (frontier and null law), S-C (hard-trim dominance), S-D (collapse lemma), and S-E (trimmed-treatment law); each computation additionally carries permanent numerical falsifiers in the repository test suite.

C Proofs for the Section 5 boundaries

Proof of Proposition 5

Work under H_0 first. The null response variance is $\operatorname{plim} \mathrm{sd}_y^2 = \beta^\top \Sigma_X \beta + \sigma_\varepsilon^2$, and Haar isotropy of β under A4a gives $\beta^\top \Sigma_X \beta = 1 + \sum_j l_j \langle \beta, q_j \rangle^2 \to 1$, since each inner product is $O_p(p^{-1/2})$; this is $v_0 = A$. For the raw floor, write $q_{\mathrm{raw}} = \|X_c^\top Y_c/n\|^2$ with $Y_c = X_c\beta + \varepsilon$ under H_0, so $\mathrm{E}[q_{\mathrm{raw}}] = \mathrm{E}[\beta^\top (X_c^\top X_c/n)^2 \beta] + \mathrm{E}\|X_c^\top \varepsilon/n\|^2$. Averaging the first term over the Haar draw of β turns it into a trace identity, $\mathrm{E}[\beta^\top \hat{\Sigma}^2 \beta] = \mathrm{tr}(\mathrm{E}[\hat{\Sigma}^2])/p = 1 + c + o(1)$, because $\mathrm{tr}(\hat{\Sigma}^2)/p \to 1 + c$ for a Wishart-type sample covariance (the Marchenko–Pastur second moment); the noise term is $\sigma_\varepsilon^2\,\mathrm{tr}(X_c^\top X_c)/n^2 = \sigma_\varepsilon^2\,\mathrm{tr}(\hat{\Sigma})/n \to \sigma_\varepsilon^2 c$. Summing, $M_0 = 1 + c(1 + \sigma_\varepsilon^2) = 1 + cA$. Under H_1 at strength g along direction dir, the numerator gains two pieces: the coherent mass enters the n-normalized cross-moment directly, contributing $g^2\omega$ riding on the spike coordinates, and the bulk leakage of the factor vector through the design's noise part, $\|U_0^\top(\gamma f)\|^2/n^2 \to cg^2$, where U_0 collects the non-spike rows of the loading basis and per-coordinate leakage variance is $c = p/n$. Hence $\mathrm{E}[q_{\mathrm{raw}} \mid g] \to M_0 + g^2(\omega + c)$. Dividing by the realized response scale (so $\mathrm{sd}_y^2 \to v_0 + g^2$ under H_1, since the raw link contributes variance g^2 before standardization) produces the displayed curve $\mathrm{E}[q \mid g] \to \{M_0 + g^2(\omega + c)\}/(v_0 + g^2)$, and algebra gives the fixed-point form with $m_0 = M_0/v_0$ and $\omega_\star = 1/A$: the shift is $\Delta(g) = -g^2(\omega_\star - \omega)/(v_0 + g^2)$, which satiates at $|\omega_\star - \omega|$ as $g \to \infty$. The born-blind condition follows by requiring the maximal standardized shift to fall inside a null standard deviations. $\square$

Proof of Proposition 4

The standardized pair has population second-moment matrix $\Omega(g) = \begin{pmatrix} 1 & \bar{u}^\top \\ \bar{u} & \Sigma_X \end{pmatrix}$, $\bar{u} = (\Sigma_X\beta + \Lambda\gamma)/\sigma_y(g)$, whose spectrum splits into the common bulk and spikes $\tau_j = 1 + l_j$ plus at most finitely many detached levels solving the secular equation (under A4a the $\Sigma_X\beta$ piece of u shifts each κ_j by only $O_p(p^{-1/2})$ and is dropped from the displayed couplings; its remainder lives in the absolutely continuous part and is absorbed by part (iii)) $\tau = 1 + \sum_{j \le r} \kappa_j/(\tau - \tau_j)$, $\kappa_j = l_j g^2 \mathrm{dir}_j^2/\sigma_y^2(g)$. This is the rank-$2r$ secular identity for a block arrowhead matrix, obtained by eliminating $(\tau I - \Sigma_X)^{-1}$ from the eigenvalue equation.

(i) Top edge. Define $f(\tau) = \tau - 1 - \sum_j \kappa_j/(\tau - \tau_j)$ on (b_+, ∞) where every $\tau - \tau_j > c + 2\sqrt{c} - l_j > 0$. Then $f'(\tau) = 1 + \sum_j \kappa_j/(\tau - \tau_j)^2 > 0$, so at most one root exists, and it exists iff $f(b_+) < 0$, i.e. $b_+ - 1 < \sum_j \frac{\kappa_j}{b_+ - \tau_j} = \frac{g^2}{A + g^2} \sum_j \frac{l_j\,\mathrm{dir}_j^2}{c + 2\sqrt{c} - l_j}$. The left side is the constant $B = c + 2\sqrt{c}$; the right side factors as $\{g^2/(A + g^2)\}\,\omega_e$ with $g^2/(A + g^2)$ increasing toward 1. Detachment at strength g therefore holds iff $\{g^2/(A + g^2)\}\,\omega_e > B$: possible at some strength iff $\omega_e > B$, and then automatic for all sufficiently large g. For scaling sub-profiles ($l_j = w\sqrt{c}$) each denominator is bounded below by $c > 0$ while numerators scale like $\sqrt{c}$, giving the computed ratios $\omega_e/B = 0.235, 0.081, 0.036$ at $c = 0.2, 0.8, 2.0$: never detached.

(ii) Bottom edge. On $(-\infty, b_-)$ every denominator is negative; $f(\tau) \to -\infty$ as $\tau \to -\infty$ while $f(\tau) \to +\infty$ as $\tau \uparrow b_-$, and $f' > 0$ throughout, so exactly one root exists for every nonzero coupling pattern, sticking to b_- as all couplings vanish. The root can sit at depth one below the edge only if the secular balance admits a solution there; clearing denominators at that evaluation point yields, for equal spikes, the wake inequality $(1 - b_-)\big((1 + l) - b_-\big) < \omega$ with saturated couplings, which fails at the three equal-spike scaling cells with margins 0.64 versus 0.22, 1.42 versus 0.45, and 1.27 versus 0.71: the bottom root never wakes.

(iii) Equal bulk laws. Inside the no-detachment region, H_0 and H_1 populations differ by a matrix of rank at most $2r + 2$; interlacing bounds the Kolmogorov distance of the two empirical spectral distributions by that rank over n, so all absolutely continuous limit laws coincide. $\square$

D Reproducibility

Every number in this paper regenerates from frozen artifacts. Table 4 collects the frozen profile constants needed to reconstruct every headline constant quoted in the text.

Each headline result whose main-text treatment is a sketch carries its complete computation in a named supplementary document, matching the descriptions above: the capture ledger (including the ridge interpolation) in S-A, the frontier and null-law closure in S-B, the dominance theorem in S-C, the collapse lemma in S-D, and the trimmed-treatment law in S-E; the $r = 1$ ledger and the two Section-5 proofs appear in Appendices A–C of this paper itself.

The simulation grids were defined before any sweep data existed and are hash-locked (`configs/grid_*.json`, grid hash `a920e554e692`); every result row carries the hash of the model card and assumption ledger pair, so no undocumented modeling knob can enter a reported cell. The global seed is 20260823, with

Object	Frozen value	Used in
Scaling sub-profile	$l_j = w_0\sqrt{c}$, $w_0 = 0.5$	Secs. 4–5
Scaling mixed profile	leading $w_0 = 2.3$–2.4 by stratum (2.375 at $\theta = \pi/6$)	Sec. 5
Link direction	$\theta = \pi/6$ unless swept	Secs. 4–5
M2 weak-injection grid	$c \in \{0.2, 0.8, 2\}$, $\|\pi\|_2^2 = 1$ sparse, $\sigma_\nu^2 = 1$	Sec. 3, 6
Treatment loading	single spike $(\delta_g, l) = (0.417, 4)$ giving middle Λ_D term 0.058	Sec. 3
Frontier strata	mixed profile, $\theta = \pi/6$, $c \in \{0.2, 0.8, 2\}$, $n = 2000$	Sec. 4
Blind strata	fully subcritical profiles or $\gamma \perp e_1$; 9 of 12 strata	Sec. 4
Benchmark aspects	six configs with $c \in [0.20, 4.64]$	Sec. 7

Table 4: Frozen grid specifications underlying the headline constants ($\Lambda_D = 1.558 = 0.5 + 0.058 + 1$, wake margins, ω_e/B ratios).

per-replication seeds derived deterministically from configuration and replication identifiers. Figures regenerate exclusively from parquet files through versioned scripts (`code/make_pilot_figures.py`, `code/make_phase2_figures.py`); no figure is edited by hand.

The identity layer is guarded by permanent tests: classical transcriptions (Marchenko–Pastur edges, BBP outlier locations, BGN overlaps, Tracy–Widom null calibration) run to stated tolerances in `tests/test_identities.py`, and each theorem carries its own falsifier suite, including `tests/test_theory_T1.py` (39 checks covering an (l, c) grid, n-scaling probes, and corners near $c = 1$) and the trimmed-law and visibility-law suites (48 further checks). The full suite is green at the submission tag. Large sweeps executed as self-contained Google Colab shards (38 phase-two notebooks plus six benchmark notebooks) with per-file checksums consolidated locally; the same shards constitute the computational appendix. The package `confounderalarm` (version 0.1.0) installs cleanly in a fresh virtual environment and its examples run end to end. An artifact manifest with sha256 checksums for every parquet, figure, and table used here accompanies the repository, and `paper/claim_map.md` links each empirical claim to its figures, artifacts, and configuration identifiers.

Data and code availability

All datasets are public: AddNeuroMed expression series GSE63060 and GSE63061 (NCBI Gene Expression Omnibus), the IHDP covariate benchmark mirror of Hill (2011), and the k401ksubs extract from Wooldridge (2016). The simulation and analysis code, the frozen grid configurations, the `confounderalarm` package, supplementary theory documents A–E, and the checksummed artifact manifest are released with the paper.

Competing interests

The author declares no competing interests.

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